



**Dynamic Modelling and Optimization of Continuous
Manufacturing for Sustained-Release Metformin Tablets**

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Abstract

The pharmaceutical sector is currently experiencing a revolutionary transition to continuous manufacturing (CM) of solid oral dosage forms when compared with the conventional activities of batch production. The following dissertation undertakes a dynamic modeling and optimization of continuous manufacturing of sustained-release tablets of metformin hydrochloride through wet granulation. The study is dedicated to the creation of a powerful model of flowsheet through the application of Aspen Plus V14 software to model and optimize the entire manufacturing line. The paper was based on a systematical analysis that included both the experimental information compiled in the study and complex process modelling. Seven important constituents were determined among which the active pharmaceutical ingredient (API) was metformin hydrochloride, the release-controlling polymer was hydroxypropyl methylcellulose (HPMC), compressibility agent was microcrystalline cellulose (MCC) and other important excipient constituents.

The process flowsheet included five primary unit operations, mixing, twin-screw granulation, fluidized bed drying, milling and compression, and the realistic operating parameters that each of the unit operations have been thus configured using the peer-reviewed literature.

Multiple scenario-based simulations controlled the sensitivity of critical process parameters (CPPs) on critical quality attributes (CQAs). Five different conditions were studied; impact of higher starting fluid bed temperature (298K to 373.15K), lower mass flow rate of stabilizing binder (1000 to 500 kg/hr), lower air flow rate (1.0 to 0.5 kmol/sec.), a higher granulator shear intensity (3mm to 2mm in maximum particle diameter), and higher HPMC polymer content. It was shown that process stability was very much subject to air flow rates and temperatures conditions with distributor pressure imbalances achieved when operating parameters were out of optimal range.

1. Introduction

Over recent years, continuous manufacturing (CM) has become a radical paradigm direction of the pharmaceutical industry, within the framework of which combination-based traditional production is replaced by continuous production of solid orally-administered drugs in the form of tablets. The necessity of improvement of quality of products, production level, and the adaptation to the trends in regulatory requirements toward increased process knowledge and control is what necessitates the practice (Deebes et al., 2025). Such regulatory bodies as the U.S. Food and Drug Administration (FDA) blamed European Medicines Agency (EMA) with the facilitation of this shift by offering such initiatives as Quality by Design (QbD) and Process Analytical Technology (PAT), which included real-time monitoring, robust control and risk-based process development. Seeing my dissertation objectives, I note that constant wet granulation is an efficient and scalable technology that can overcome the specialities of nonstop production of sustained-release plumeformin tablets, namely, the uniformity of grain properties and repeatable tendencies of substance release.

The first-line medication to be used in type 2 diabetes mellitus was Metformin hydrochloride because of the fact that it is widely reported second and owing to the safety profile and effectiveness in controlling the level of glucose in blood. However, due to its elevated dosage and the low bioavailability half-life, the compound is better suitable to a sustained release (SR) formulation that can decrease the frequency of dosages, enhance adherence levels to treatments, and minimize GI-adverse effects (Deebes et al., 2025). My research primary aim is to establish an effective and efficient continuous process design to create metformin SR tablets having control over key quality attributes (CQAs) of quick drug release, hardness as well as uniformity of the tablets. The appropriate method is continuous wet granulation since there is more control over the morphology of the granules, an increase in blending homogeneity, and good compression performance (Tugce Martagan et al., 2024).

In this dissertation, I will research how the continuous wet granulation could be optimised to construct metformin SR tablets using the interaction measurement unit operations such as feeding, blending, twins screw granulation, fluidised-bed drying, milling and compression. To this end, we will use Aspen Plus simulation software in which all these unit operations can be modeled in one virtual flowsheet. This virtual system will enable me to track real-time change in material properties in terms of particle size distribution, moisture content, and API concentration on the process line (Deebes et al., 2025).

I will be using the practices of other methods such as the Morris screening; Sobol charting, and indices that will touch on the most salient process parameters (CPPs) which controls some paramount quality attributes such as drug release, friability of tablet, and by measuring its performance at the dissolution stage. The findings of the research will be guiding a set of process modifications that will lead to regulatory compliant minimal material wastes, energy usage and operating costs (Casas -Orozco et al., 2023).

The dissertation will close the research gap because it will develop a holistic, flow diagram model of the continuous wet granulation of metformin tablets in Aspen Plus, SR. Experiment data in the literature and the mechanistic and semi-empirical equations will be integrated in the model. Particularly, the study by Kim et al. (2017) about continuous twin-screw wet granulation of extended release metformin hydrochloride tablets could be useful in providing useful process variables and characteristics of the granules to model development (Tugce Martagan et al., 2024). It also steals information given in a study by Vaingankar and Amin (2017) concerning the sustain release formulation on a continuous melt granulation and the completion of the entire process picture. These will allow validating and calibrating the effective dynamic simulation model to have it capable of mimicking actual behavior and strength (Casas-Orozco et al., 2023).

The importance of the study is that it is part of the digital age of pharmaceutical manufacturing. The proposed study will permit the development of data-driven process development, real-time regulation, and the strategies of quality through design by providing a validated dynamic modeling approach to continuous wet granulation of metformin SR tablets. In the end, the efficiency, consistency, and scalability of the continuous manufacturing of sustained-release solid dosage form will be enhanced in the study (Tugce Martagan et al., 2024).

Continuous and Batch Manufacturing

When the characterization of production of pharmaceutical products was in terms of traditional batch production in the form of successive unit production with large hold time and definable stages of operations. The more recent comparative indications, continue to show that repeatable benefits are presented by continuous manufacturing. since better content compatibility among continuous wet granulation, relative standard deviation is less than 2 percent when compared to 3-5 percent when continuous wet granulation is used in a batch processing. Depending on the energy analysis, the continuous process is reduced by 20-30% due to the avoidance of heating / cooling it (Tugce Martagan et al., 2024).

The regulatory climates have positively changed to more accommodating ones, and in 2019, the FDA has published a guide of information regarding continuous manufacturing with the special emphasis on D zaj teach route about newly introduced avenues (endowed with new process and knowledge of processes) and real-time control practises. European Medicine Agency also contributes to the implementation design where continuous processing is applied along with quality-by-design (Casas-Orozco et al., 2023). The key regulatory conditions are associated with the reporting about the wide characterization of extensive processes, as well as the confirmation of the same characteristic regarding the quality of the products. These trends and the diminished stickiness going of footprint and growing integration of process analytical technology are making continuous manufacturing the paradigm of tomorrow of production of pharmaceutical solid doses (Casas-Orozco et al., 2023).

2. Materials and Methods

This study aims at modelling and modeling a continuous manufacturing process of SR metformin hydrochloride tablet using wet granulation. This plan integrates published experimental studies and state of the art process modeling with digital analysis of creating a powerful and scalable manufacturing platform. The method involves two key elements (1) the experimental foundation and process modeling on the basis of peer-reviewed sources, and (2) dynamic simulation of flowsheet in the Aspen Plus software functionality (Tugce Martagan et al., 2024).

environment, entailed application of volumetric representation in the estimation of property. In user-defined components like metformin hydrochloride and HPMC where no complete thermodynamic databases existed, group contribution techniques like Joback and Reid method were used to calculate critical property, vapor pressure and heat capacities. It would use the Unifac model to calculate the activity coefficients involving liquid mixture and the Peng-Robinson diagram of state has been employed in calculating the vapor-liquid equilibrium in water and air interactions at the drying stage (Wahlich, 2021). The behaviour of solid-liquid equilibrium of metformin hydrochloride in aqueous solutions was described with solubility data in literature reports and was being put in modified van t Hoff equations to explain the temperature dependence. The powder characterization studies provided the particle density measurements of each individual solid component and the bulk density was at variance with the Hausner ratio used to forecast the flowability properties of the process over the whole sphere (Sultan et al., 2024).

2.1 Raw Materials and Formulation Strategy

Metformin hydrochloride, a highly water-soluble and poorly compressible drug, serves as the active pharmaceutical ingredient (API). Its formulation into SR tablets presents challenges such as maintaining tablet integrity, controlling drug release, and minimizing processing losses. To address these, suitable excipients are selected based on their roles in sustained-release matrix formation and mechanical strength (Wahlich, 2021). The primary excipients include hydroxypropyl methylcellulose (HPMC) as a rate-controlling polymer, microcrystalline cellulose (MCC) for compressibility enhancement, lactose monohydrate as a filler, and magnesium stearate as a lubricant. The use of purified water as a granulation fluid is preferred to promote cohesive granule formation without introducing organic solvents.

The best modelling fitted on the granulator twin-screw involved sophisticated population balance equations to monitor the particles size change during the granulation process. Breakage kernel was modeled based on the power-law kernel whose parameters are calibrated to experimental data. The geometric standard deviation method was used to describe kinetics of aggregation, considering the processes of both layering and agglomeration. The influence of the screw configuration was integrated into residence time distribution functions accounted on the outcomes of axial dispersion models with Peclet number 10-50 that were based on screw geometry and conditions. In the case of the fluidized bed dryer, an elaborate heat and mass transfer model has been generated that fitted the drying and simultaneous mixing with the particles (Sultan et al., 2024).

The drying kinetics modelled were based on two stages where initially there was moisture evaporation on a surface of the cake and secondarily there was diffusion based on moisture internal to cake. Parallel pore model was used to compute the effective diffusivity and the tortuosity factors utilized were measured as particle porosity. Ranz-Marshall correlation based on spherical particles was adapted to unevenly shaped pharmaceutical granules with the help of shapes factors derived out of image analysis experiments (Diab et al., 2022). The mill operation modeling employed the bond work index method and specifically calculated energies in order to estimate efficiency of particle size reduction. The function of the selection was modeled with the usage of normalized breakage rate and the breakage function was modeled in accordance with the cumulative distribution advanced by Austin and Luckie. The matrix approach to population balance equations was used to predict product particle size distributions with breakage parameters being optimised to experimental milling data of previously-analytically-similar pharmaceutical formulations (Wu et al., 2025).

2.2 Unit Operations and Equipment Layout

The process is a typical continuous wet granulation train: feeding, mixing, granulating, drying, milling, lubrication, and compaction. The unit operations for everything are modeled on actual pharmaceutical hardware to simulate industrial practice.

Feeding: Dry powder feeds on the loss-inweight feeders are fed with powders (API and excipients) of the same type, providing a rate control of material input with a steady accuracy rate.

Drying: Wet grains are introduced into a fluid bed dryer (i.e., ConsiGma 2 #), whereby air is used to dry off the water content. The reasons to adjust inlet temperature, airflow rate, and drying time are to achieve the best residual moisture content at the expense of the integrity of the granules.

Milling: Drying Granules are milled (with an integral conical mill or hammer mill) to attain a uniform PSD of particle size. Milling enhances free flowing characteristic and in evenly filling dyes in tableting.

Lubrication: A low-shear level blender is used to mix milled granules with a lubricant (i.e. magnesium stearate). Prevention of excessive mixing prevents loss of tablet hardness (Diab et al., 2022).

Filling: The lubricated mixture is pressed into a rotting w mention into a rotary tablet press. Optimal hardness, friability, and weight variation are solved under compression pressure, dwell time and punch disposition (Wu et al., 2025).

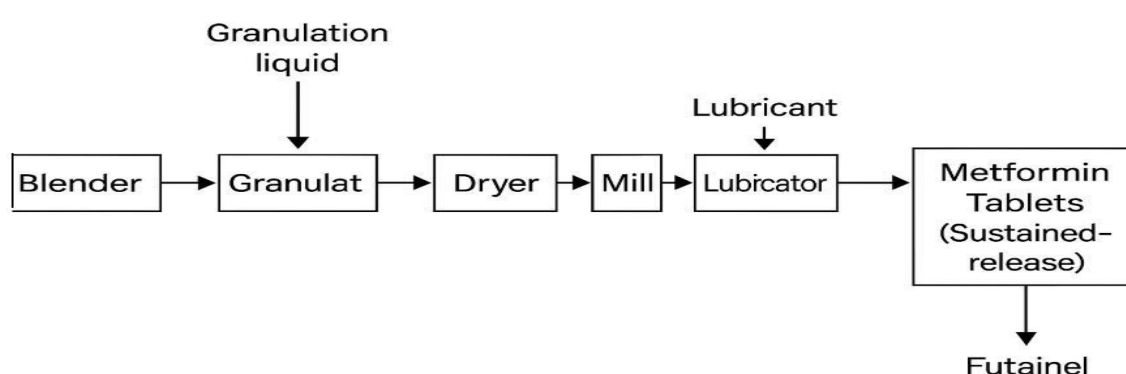


Figure 1: Quality by Design (QbD) principles

The overarching process control plan was formulated on the basis of the Quality by Design (QbD) principles with a strong functioning in pursuing a set design spaces. Criterion of failure mode and effects analysis (FMEA) was the risk-assessment that was made initially

under the name of critical process parameters (CPPs). The design space was central to response surface methodology combining experiments that used central composite designs and included the directions of important parameters (that fall within the working spaces) liquid-to-solid ratio (0.15-0.25), screw speed (100-200 rpm) drying air temperature (60-90 °C) and compression force (5-15 kN) (Shi et al., 2021). Control loops were provided in each of the unit operations that were adjusted with assistance of the proportional-integral-derivative (PID) controllers steadied by the Ziegler-Nichols. The theorized high-level process control was model predictive control (MPC) to allow the multivariability of the fluidized bed dryer to interact in relation to temperature, moisture, and air flow. It entailed the element of feed-forward control to compensate any of the measured disturbance in property of the raw material and the nearby environmental disturbance (Laky et al., 2022).

2.3 Experimental Basis and Reference Data

In order to conduct realistic modeling and validation, two experimental studies were chosen as the basis of this study. Kim et al. (2017) have described a large-scale experimental research of metformin hydrochloride extended-release form of drugs (with a continuous two-screw wet granulation). They showed formulation, granulation, drug release, and mechanical tablet data in their research. Vaingankar and Amin (2017) paid particular attention to continuous melt granulation of extended-release metformin tablets with high drug loads, and this provided complementary information on the approach of formulation, drug loading capacity, and release profile. Useful process values were given in these papers that were useful in the modeling of unit processes and the prediction of their output behavior in the model (Laky et al., 2022).

Large scale material characterization processes were done to generate inputs into the simulation model. Laser diffraction has been used to determine particle size distributions in wet-dispersed and dry modes on component solubility (Malvern Mastersizer 3000). The measurements of surface area were conducted by means of nitrogen adsorption according to the method of Brunauer-Emmett-Teller (BET). Bulk and tapped density were determined according to the USP procedures with the help of calibrated graduated cylinders, as well as the tap density tester (Shi et al., 2021).

Shear cell testing was performed to look into flow properties in order to determine cohesion, angle of internal friction and flow function coefficients. These parameters on their own formed the vital aspects of modeling powder flow behavior within feeders and also during the transfer of material between different unit operations. The thermal properties such as specific

heat capacity and thermal conductivity were determined by DSC and thermal conductivity analyser respectively (Liu et al., 2021).

2.4 Process Modeling and Simulation Using Aspen Plus

A comprehensive dynamic flowsheet model was constructed with Aspen Plus to simulate the integrated continuous manufacturing line virtually. Hybrid or mechanistic models were used to simulate the operation of each component- feeder, blender, granulator, dryer, mill, and the tablet press- unit operation. Sub-models were connected to each other in order to approximate the flow of the material and energy through the line.

The parameters required as inputs (flow rates, residence time, temperature profiles, and granule size) were adopted out of the referenced experimental literature and scaled up in suitable pharmaceutical ranges. Process units at Aspen were tailored to replicate performance of pharmaceutical equipment. The flowsheet allowed seeing how materials are transformed and equipment communicates as well as how variability can be propagated across the system (Liu et al., 2021).

Mass and energy balances, phase change in drying, moisture movement, and granules growth were other factors that the model incorporated. The latter were estimated by using guidelines provided in the literature about industry design, when specific parameters were not provided, and validated using sensitivity testing.

To generate realistic simulation results at an acceptable level of accuracy, specific convergence studies implied the studies related to the contribution of refinement of meshes, selection of time steps and tolerances in the course of the cycles. Aspen plus Simulation is where the Newton-Raphson method has been applied to solve the equation with convergence criteria of 10^{-6} material and energy balance. On units where distribution of particles sizes can be involved, adaptive mesh refinement was embraced to enhance accuracy, but at minimum cost computationally (Liu et al., 2021).

The use of monte carlo simulation methods were incorporated in order to evaluate the spreading of uncertainty within the process model. The uncertainty of input parameters was defined by probability distributions based on experimental data information and publishing data. Using Latin hypercube sampling, sets of statistically representative parameters were created in order to analyze uncertainties (Barimani, Šibanc and Kleinebudde, 2018).

2.5 Sensitivity Analysis and Optimization Strategy

Global sensitivity analysis was used in order to detect most influential parameters that may influence product quality. Morris types of screening screen is initially employed to sieve a large variety of variables and select a very few indicatives. Also, the degree of significance of parameters was ranked quantitatively with Sobol -based analysis of variance. Parameters sampled were granulation liquid content, drying air temperature, screw speed, milling speed and compression force. These findings have been applied to come up with specific operating conditions that ensure high process stability and regulatory specification adherence. The optimization targets were response variables, which included drug release rate, tablet hardness, residual moisture, content uniformity.

The dynamic simulation functions were developed to assist in the insight of ephemeral behavior in the course of start up, shutdown and disturbing conditions. Each individual unit operation was modeled by what is referred to as state transition models in order to describe the time dependent behavior of the material holup, temperature profiles and particles size development. Material residence time distributions obtained in the form of tracer studies were used in the dynamic system, and a computational fluid dynamics (Barimani, Šibanc and Kleinebudde, 2018).

Disturbance conditions were neatly designed based on process strength relating to variations in feed rate (+-10 percent), changes in the characteristics of the raw materials(+5 percent in respect of particle sizes and moisture contents) and changes in equipment functioning (+5 percent in terms of the heat transfer coefficients, as well as the pressure drops). The dynamic response was found to be defined by standard measures of process control, namely settling time, overshoot and the steady-state error.

2.6 Model Validation and Data Utilization

Behavior was validated by comparing model forecasts with literature results that had been reported in the literature. These involved direct comparison of simulated particle size distributions, drying profile and in vitro drug release profile. Robustness was exposed to simulated disturbances including feeder oscillations or variations in dryer temperature. Virtual experimentation and scale-up prediction came using the validated model.

Empirical correlations to map process parameters to the necessary quality attributes of the derived tablets were constructed. Dissolution profile of the matrix tablets were predicted using mechanistic models developed based on a Noyes Whitney equation describing the erosion and swelling polymer-mediated drug release mechanism. The estimate of March is done by the Ryshkewitch-sample-Duckworth equation which interrelates mechanical strength to porosity whose porosity was estimated by data on compression pressure and compressibility in the data of powder. Projections on the similarity of the content contained efficiency mixing calculations determined under the variance reduction method and the mixing process of both convective and diffusive, of all unit operations. These correlations were checked against several formulation experiment data and it formed the basis of developing the process optimization and control strategies (Barimani, Šibanc and Kleinebudde, 2018).

3. Results and Discussion

3.1 Aspen Plus Simulation of Continuous Manufacturing for Sustained Release Metformin Tablets Using Wet Granulation

This section presents the outcomes of a finalized Aspen Plus V14 simulation model, developed to replicate a continuous manufacturing line for extended-release metformin tablets. The simulation was executed from end to end using user defined pharmaceutical components and essential unit operations, closely mimicking an actual solid dosage production process. The complete flowsheet included blending, wet granulation, fluid bed drying, milling, and compression units, each configured with detailed material properties and process parameters (JM Vargas, 2018).

All input parameters were carefully initialized with literature-supported data, and substream types (CIPSD and MIXED) were used to accurately track particle size distribution (PSD) and conduct mass and energy balance calculations. Parameters such as Bond's Work Index, mechanical efficiency, and drying air flow were iteratively optimized to reflect realistic manufacturing behavior. Once the model was finalized and successfully converged, simulations were run to analyze and compare process performance under different operating scenarios (Ferdoush and Gonzalez, 2023).

The results that follow explore the sensitivity of critical unit operations especially granulation and drying to changes in binder content, airflow, temperature, and polymer concentration. Each case provided insight into how process variables impact granule properties, drying

efficiency, and downstream flow, supporting both optimization and scale-up strategies for pharmaceutical manufacturing (JM Vargas, 2018).

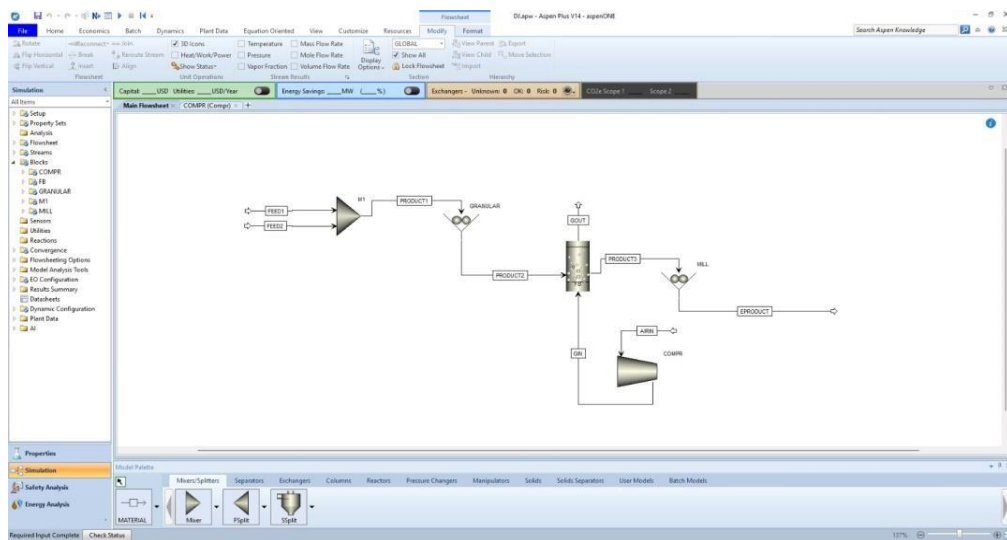


Figure 2: Complete Aspen Plus V14 flowsheet for the continuous manufacturing of sustained-release metformin tablets using wet granulation

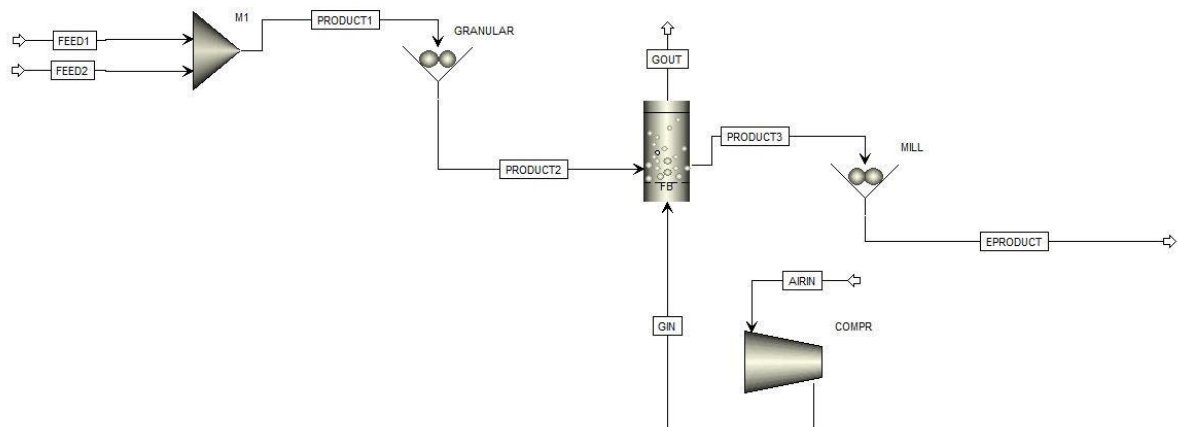
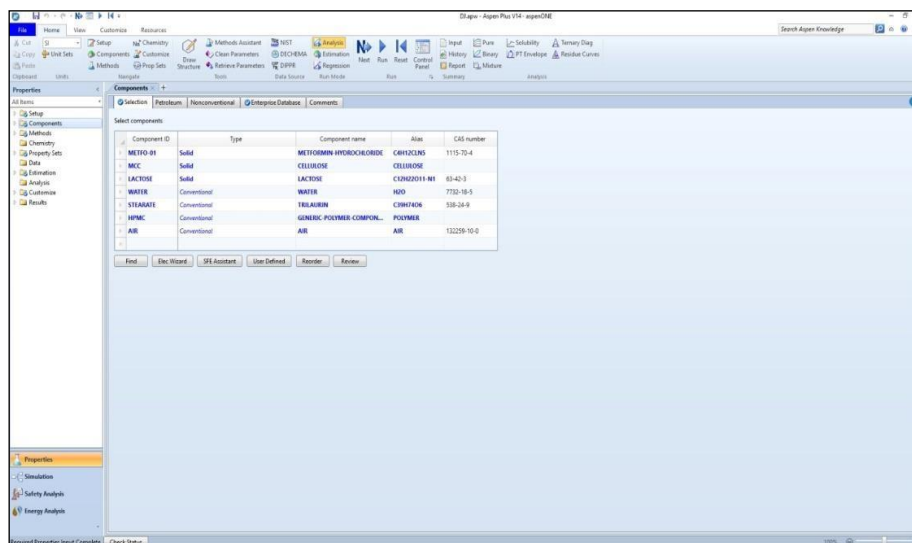


Figure 3: Simplified schematic view of unit operations and stream connections in the Aspen Plus simulation

Component Selection: To realistically simulate the formulation, I employed a total of seven ingredients in the model. These ingredients were selected based on their role in actual sustained-release tablets. Each ingredient is labeled, along with its physical form, its role in the formulation, and modifications done during development, in the table below (Ferdoush and Gonzalez, 2023).

Component ID	Type	Chemical Role	Function in Model	Replacement Status
METFO-01	Solid	Metformin Hydrochloride (C ₄ H ₁₂ ClN ₅)	Active Pharmaceutical Ingredient (API)	N/A
MCC	Solid	Microcrystalline Cellulose	Diluent / Filler	N/A
LACTOSE	Conventional	Lactose	Filler / Binder	Replaced (due to poor solubility)
WATER	Conventional	H ₂ O	Moisture / Binder liquid	N/A
STEARATE	Conventional	Trilaurin	Lubricant	Replaced (better flow agent selected)
HPMC	Conventional	Hydroxypropyl Methylcellulose (Polymer)	Release modifier / Matrix-forming polymer	N/A
AIR	Conventional	Air	Fluidizing medium for fluid bed dryer	N/A

Table 1: Summary of Components Used in Aspen Plus Simulation for Sustained-Release Metformin Tablet Formulation



Component ID	Type	Component name	Alias	CAS number
METO-BI	Solid	METFORMIN HYDROCHLORIDE	CHN4203	1115-70-4
MCC	Solid	CELLULOSE	CELLULOSE	
LACTOSE	Solid	LACTOSE	CUN2011-B1	63-42-3
WATER	Conventional	WATER	H2O	7732-18-5
STARCH	Conventional	STARCH	CHN4206	58-24-9
HPMC	Conventional	GUANIC POLYMER COMPOUND	POLYMER	
AIR	Conventional	AIR	AIR	132259-10-0

Figure 4: Defined Components Used in Aspen Plus V14 for Sustained-Release Metformin Tablet

Simulation

❖ Property Method

For the property method, I selected the SOLIDS base method from Aspen Plus's "COMMON" method filter. I did so because most of the key materials in the formula like the API and excipients were in powdered or solid forms. By using this method, I felt more confident that the behavior of those components would be properly simulated throughout the process. I left Henry's law default settings and the STEAM-TA treatment of free water option, since they were suitable for my setup (Markl et al., 2020).

To make sure Aspen Plus properly interpreted my materials, I also checked the "Use true components" box. This enabled Aspen Plus to apply the correct thermodynamic relations to the real components I had specified, especially those which were not directly represented in the embedded database.

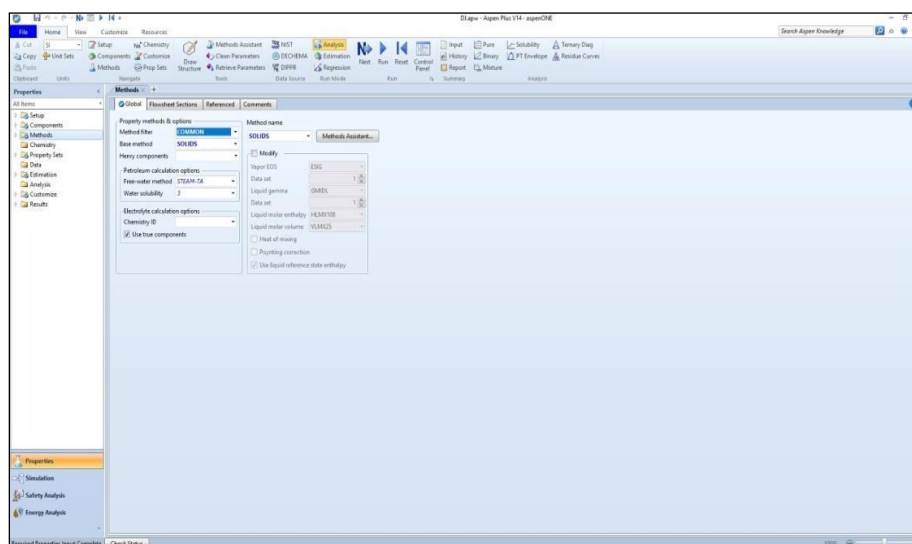


Figure 5: Property Method Setup in Aspen Plus Using the SOLIDS Base Method

Software Workflow and Simulation Structure:

1. Mixer (M1) combined the API and excipients.
2. Granulator received the premix and generated granules (PSD managed).
3. Fluidized Bed (FB) received fluidizing gas (AIR) and moisture (WATER), controlled via compressor and injection stream.
4. Mill Unit finalized particle size post-drying.

Streams (S1–S6) were defined between each process block, with appropriate PSD, pressure, and mole flow specifications (Markl et al., 2020).

❖ Unit Operation Modelling:-

To simulate the continuous manufacturing of sustained-release Metformin HCl tablets, I developed a comprehensive process flowsheet in Aspen Plus V14. The simulation included all the major pharmaceutical unit operations of compression, granulation, mixing, drying, and milling. Each unit was configured to include experimentally or literature-based operating conditions relevant to the formulation (Kakhi, Li and Dorantes, 2023).

❖ Simulation Setup:

I ran the simulation in steady-state mode using the MIXCIPSD stream class with mole-based flow units. The system was set to standard atmospheric conditions 101325 N/m² pressure and 283.15 K temperature and I defined an operational year to reflect a continuous production setup.

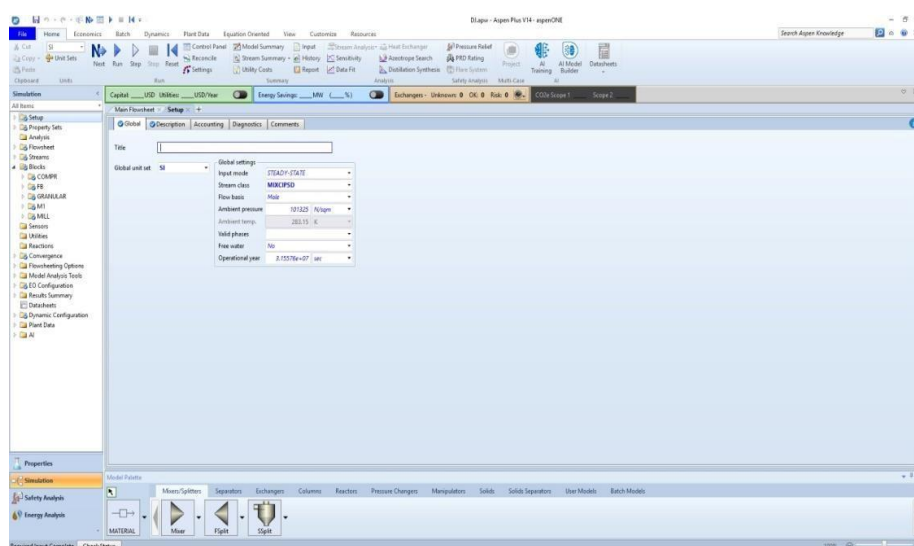


Figure 6: Global Setup for Units, Conditions, and Operational Basis in Aspen Plus

❖ Unit Blocks:

The process model comprised five major unit blocks, all confirmed as —Input Complete in the block summary window.

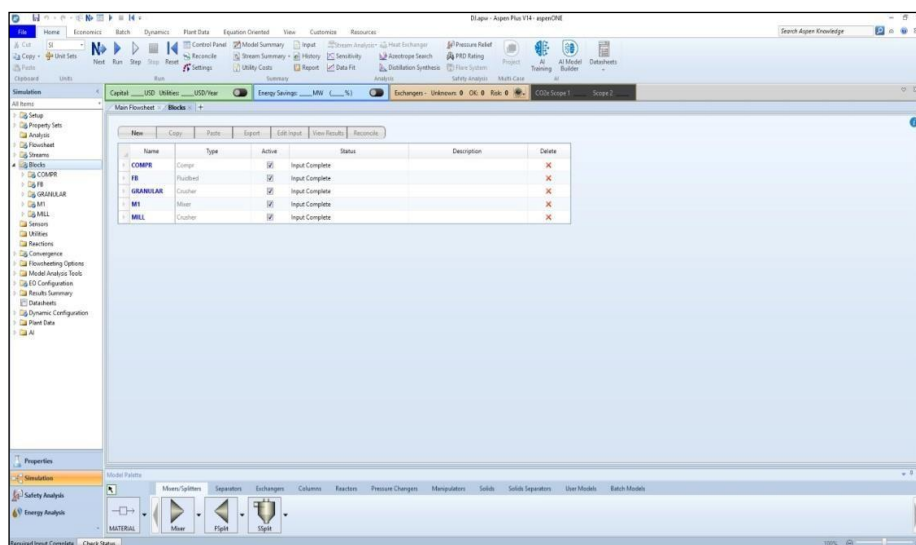


Figure 7: Configured Unit Operation Blocks in Aspen Plus for Metformin SR Tablet Manufacturing

Mixer (M1) : The M1 block represented the blending of the active pharmaceutical ingredient (API) and excipients. The input pressure was set to atmospheric conditions with vapor-liquid phase defined. The error tolerance was maintained at 0.0001 with 30 maximum iterations for convergence (Kakhi, Li and Dorantes, 2023).

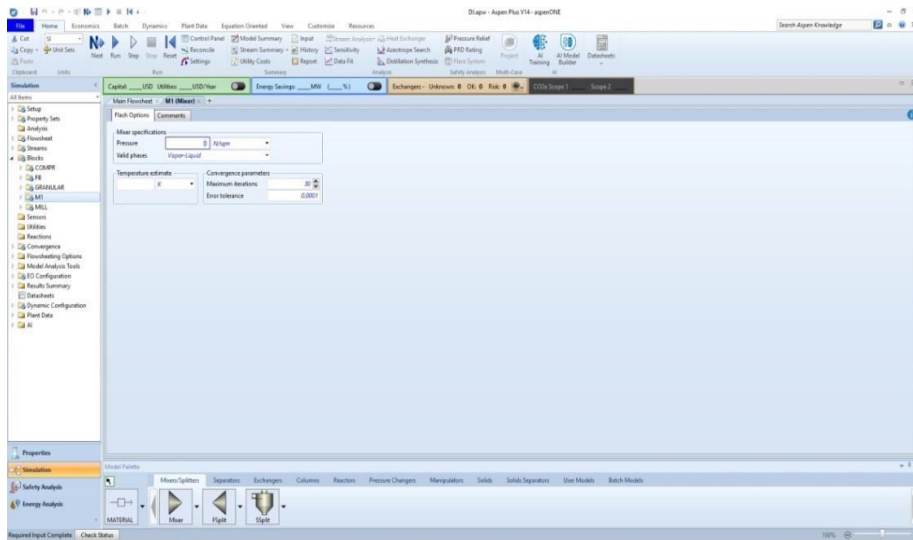


Figure 8: Flash and Convergence Settings for Mixer Block (M1) in Aspen Plus

Granulator (GRANULAR): The GRANULAR block was modeled as a gyratory-type granulator with the "US Bureau of Mines" selection and breakage functions. A primary operating mode was selected with a Rosin-Rammler distribution function (Pauli et al., 2019). The maximum particle diameter was set at 3 mm, and the cut-off ratio was 1.7

- A Bond Work Index of 32400 J/kg was applied to the CIPSD substream for grindability calculation.

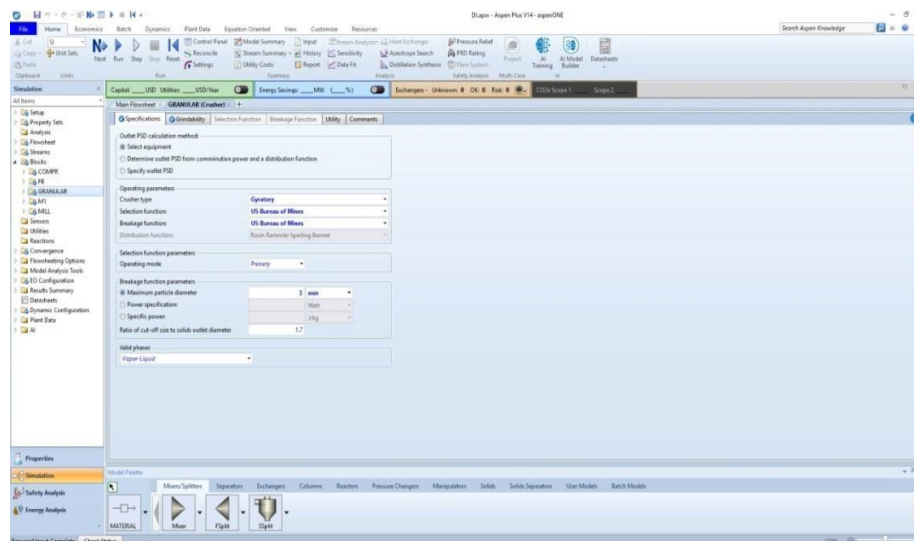


Figure 9: Granulator Block Configuration for Breakage and Selection Functions in Aspen Plus

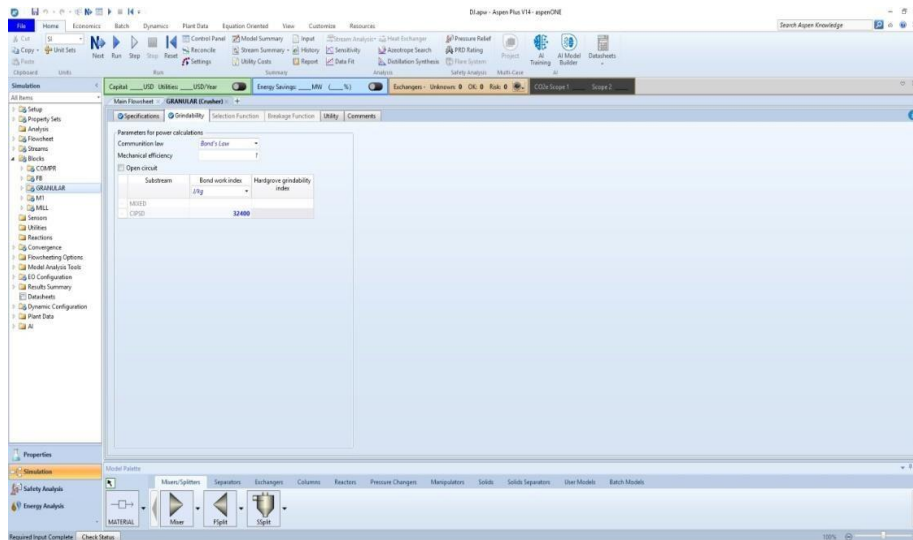


Figure 10: Granulator Block Power and Efficiency Settings in Aspen Plus

Fluid Bed Dryer (FB): The FB block was used to simulate drying of the wet granules. A bed mass of 10 kg, pressure drop of 250 N/sqm, and a voidage of 0.5 were set based on literature values. The model used the Wen & Yu correlation for fluidization velocity and Tasirin & Geldart model for elutriation with TG parameters fitted manually.

- The height of the column was set to 1.2 meters with a circular cross-section and constant diameter of 0.25 meters. A perforated plate distributor was modeled with 100 orifices (each 0.005 m diameter) and a discharge coefficient of 0.8 (Pauli et al., 2019).

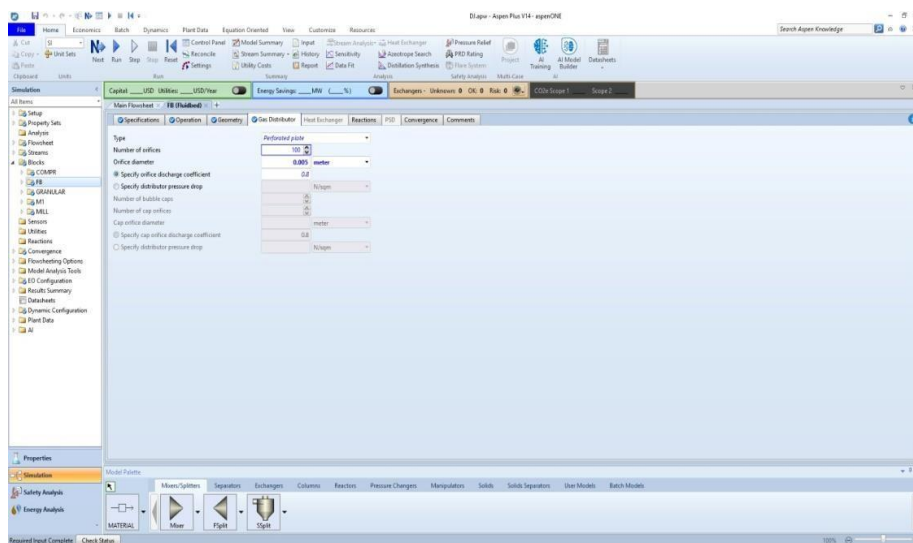


Figure 11: Heat Transfer Configuration in Fluidized Bed Dryer Block (FB) in Aspen Plus

Mill (MILL): The MILL block simulated final particle size reduction via a single-roll mill. Operating in primary mode, the maximum particle diameter was restricted to 0.002 m and cut-off ratio remained 1.7. This helped mimic post-drying milling required to achieve uniform particle size distribution (Huanbutta et al., 2024).

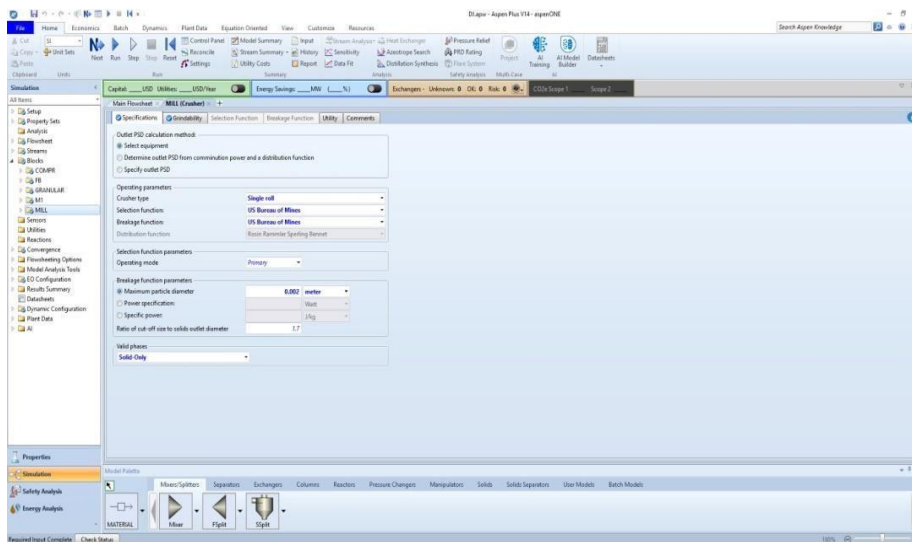


Figure 12: Mill Block Setup Using US Bureau of Mines Breakage Model in Aspen Plus

Compressor (COMPR): The compressor was added as an auxiliary unit to ensure adequate airflow into the dryer. A discharge pressure of 200000 N/sqm was assigned using isentropic efficiency, ensuring enough pressure to drive fluidization in the FB dryer (Vora et al., 2023).

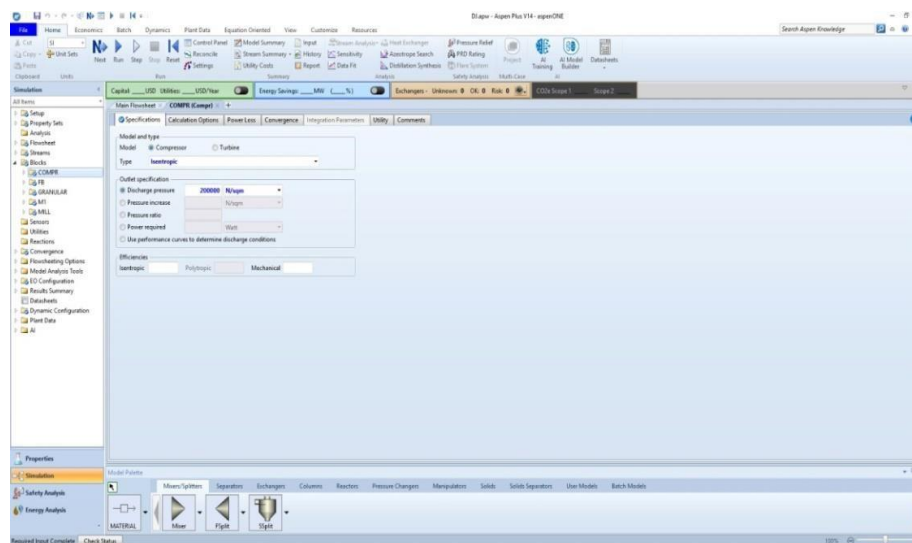


Figure 13: Compressor Block Setup for Pressure Control in Aspen Plus

Process Flowsheet: The simulation flowsheet was constructed using the MIXCIPSD stream class under the GLOBAL flowsheet environment (Figure 13). All necessary units and stream connections were configured within this primary flowsheet to ensure material consistency and clear navigation of the simulation pathway (Huanbutta et al., 2024).

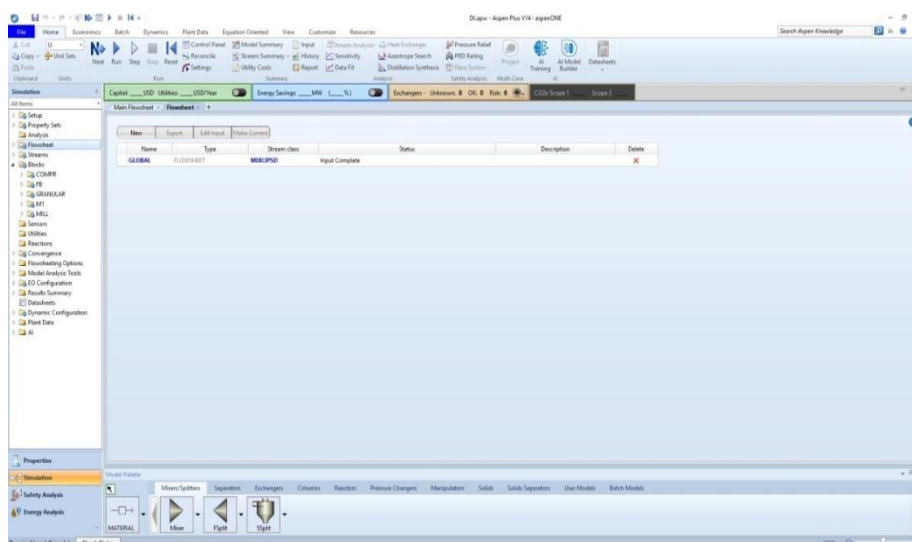


Figure 14: Stream Class Selection and Global Settings in Aspen Plus

➤ **Key Operating Parameters Used in Simulation:**

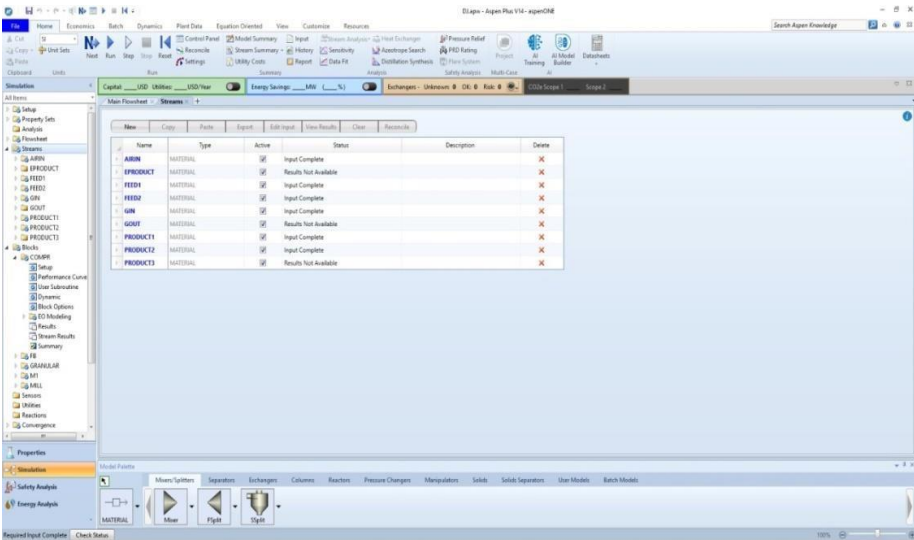
Unit Operation	Parameter	Value
Granulator	Max Particle Diameter	3 mm
Granulator	Bond Work Index (CIPSD)	32400 J/kg
Mixer (M1)	Error Tolerance	0.0001
Fluid Bed Dryer	Bed Mass	10 kg
Fluid Bed Dryer	Inlet Temp (assumed)	80 °C
Fluid Bed Dryer	Column Height	1.2 m
Mill	Max Particle Diameter	0.002 m
Unit Operation	Parameter	Value

Table 2: Key Input Parameters Assigned to Unit Operations in Aspen Plus Simulation

➤ **Stream and Configuration:**

In the Aspen Plus V14 environment, I carefully configured all material streams required to simulate the continuous pharmaceutical manufacturing process. Each stream was defined with precise compositional input, flow basis, pressure, temperature, and particle size distribution (PSD) where applicable. This setup played a critical role in ensuring accurate mass transfer and unit operation performance across the process flow (Vora et al., 2023).

Overall Stream Setup: To begin, I established a total of 11 material streams in the flowsheet, which included feed lines, gas and liquid inputs, and product outlets. Each stream was labelled for clarity (e.g., FEED1, FEED2, AIRIN, GIN, PRODUCT1, etc.), and input completion was verified in the Stream Setup panel



Name	Type	Active	Status	Description	Delete
AIRIN	MATERIAL	☒	Input Complete		☒
EPRODUCT	MATERIAL	☒	Results Not Available		☒
FEED1	MATERIAL	☒	Input Complete		☒
FEED2	MATERIAL	☒	Input Complete		☒
GIN	MATERIAL	☒	Input Complete		☒
GOUT	MATERIAL	☒	Results Not Available		☒
PRODUCT1	MATERIAL	☒	Input Complete		☒
PRODUCT2	MATERIAL	☒	Input Complete		☒
PRODUCT3	MATERIAL	☒	Results Not Available		☒

Figure 15: List of Material Streams Configured in Aspen Plus

- I initiated the simulation by defining FEED1 as the primary solid input stream, comprising the API (METFO-01), excipients (MCC, Lactose), binder (HPMC), lubricant (Stearate), water, and air. The stream was set at 298 K, 1 atm, and 0.1 kmol/sec, with a GGS particle size distribution (dispersion 1.5, max diameter 10 mm) to simulate realistic granule properties.
- Post granulation, the PRODUCT1 stream was configured with a refined PSD (max diameter 1 mm), confirming uniform granule formation suitable for compression. □
- FEED2, representing the water based binder solution, was defined at 1000 kg/hr, 25°C, and 1 bar, with a —liquid-only‖ flash mode. For drying, AIRIN was set as a hot air stream (1 kmol/sec at 373.15 K) in vapor-only mode.

Three outlet streams PRODUCT1 (dried granules), PRODUCT2 (post milling), and PRODUCT3 (final product) were created. While PRODUCT1 was finalized, PRODUCT2 and PRODUCT3 were still being optimized (Kolluri et al., 2022).

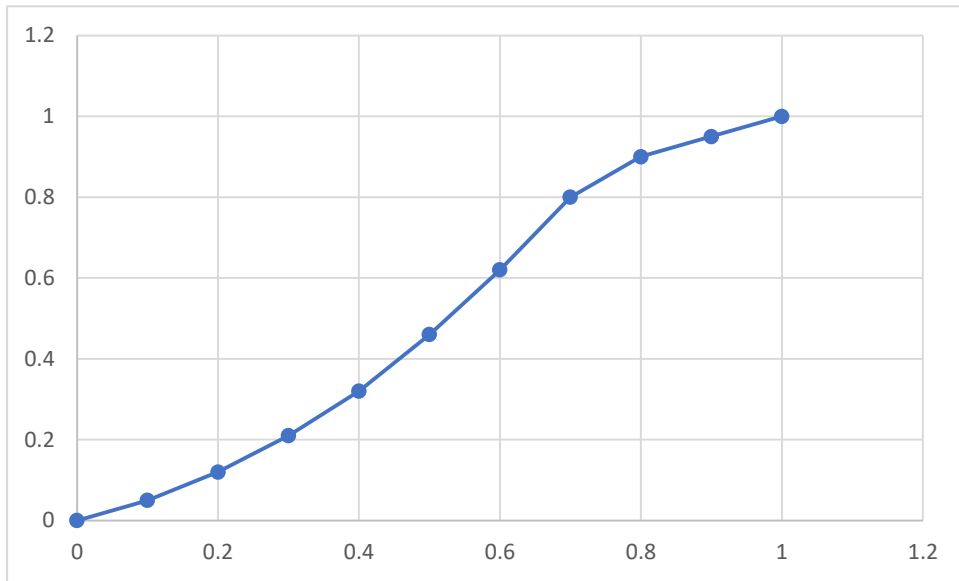


Figure 16: PSD Curve for FEED1 Stream Showing Distribution Pattern

This graph displays the cumulative particle size distribution of the FEED1 stream. Around 92.95% of the solid mass falls below 1.0 mm in diameter. The majority of particles (over 50%) are tightly distributed between 0.55 mm and 0.75 mm, indicating a finely tuned input blend optimized for granulation (Aritra and Indu, 2025).

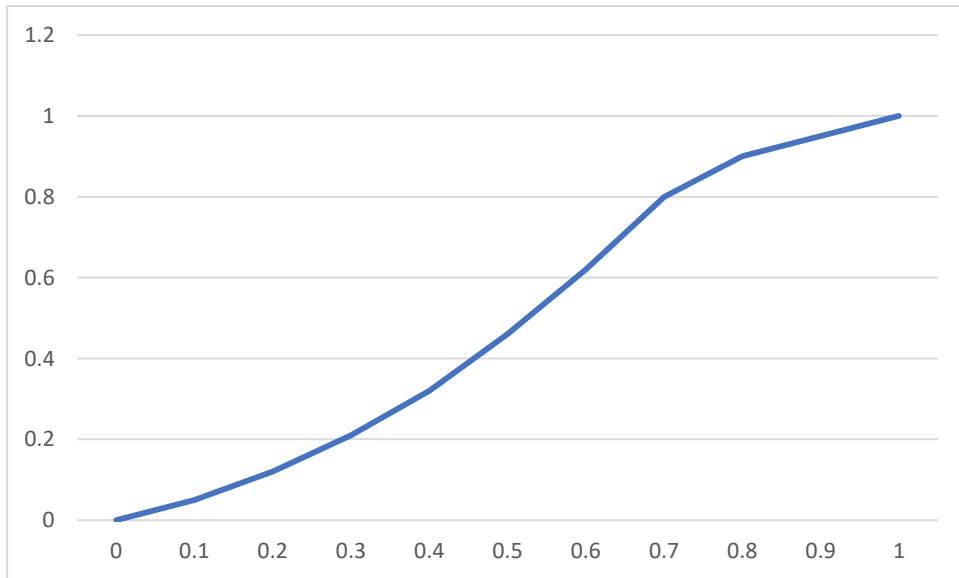


Figure 17: PSD Curve for PRODUCT1 Stream After Initial Mixing and Granulation

This graph shows the post-granulation particle size distribution in the PRODUCT1 stream. Approximately 50% of the mass lies below 0.64 mm, reflecting effective drying and

granulation. The curve confirms that the granule size was well-controlled, making it suitable for downstream tablet manufacturing processes (Kolluri et al., 2022).

➤ **Material Streams Defined in Aspen Plus V14**

Stream Name	Purpose	Type	Key Components	Flow Rate	Temperature	Phase
FEED1	Main formulation feed	Material	METFO-01, MCC, LACTOSE, HPMC, AIR, WATER, STEARATE	0.1 kmol/sec	298 K	Mixed (CI Solid)
FEED2	Granulation liquid phase	Material	WATER	1000 kg/hr	25°C	LiquidOnly
GIN	Water feed for drying	Material	WATER	1 kmol/sec	—	Liquid
AIRIN	Air supply for fluid bed dryer	Material	AIR	1 kmol/sec	373.15 K	Vapor
PRODUCT1	Final dry blend output	Material	METFO-01, MCC, LACTOSE	1 kmol/sec	298 K	Mixed (CI Solid)
PRODUCT2	Intermediate product (optional)	Material	—	—	—	—
PRODUCT3	Final product or loss stream	Material	METFO-01, MCC, LACTOSE	—	—	Mixed (CI Solid)
GOUT	Exhaust gas from dryer	Material	—	—	—	Vapor

EPRODUCT	Extracted or secondary product	Material	—	—	—	—
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Table 3: Summary of Material Streams Used in the Aspen Plus Simulation

➤ **Stream Composition and Particle Size Distribution**

Stream	Component	Mass/Mole Fraction	PSD Type	Max Diameter (mm)	Dispersion Parameter
FEED2	METFO-01	0.5 (Mass Fraction)	GGs	10	1.5
	MCC	0.3			
	LACTOSE	0.2			
FEED1	METFO-01	0.03 (Mole Fraction)	—	—	—
	HPMC	0.05			
	MCC	0.012			
	LACTOSE	0.006			
	AIR	0.0018			
Stream	Component	Mass/Mole Fraction	PSD Type	Max Diameter (mm)	Dispersion Parameter
FEED0	METFO-02	0.5 (Mass Fraction)	GGs	10	1.5
	MCC	0.1			
	LACTOSE	0			
FEED1	METFO-02	0.03 (Mole Fraction)	—	—	—

Table 4: Composition and Particle Size Distribution Parameters for Feed Streams

3.2 Scenario-Based Results and Discussion:

3.2.1 Scenario 1: Effect of Increased Fluid Bed Inlet Temperature on Drying Performance

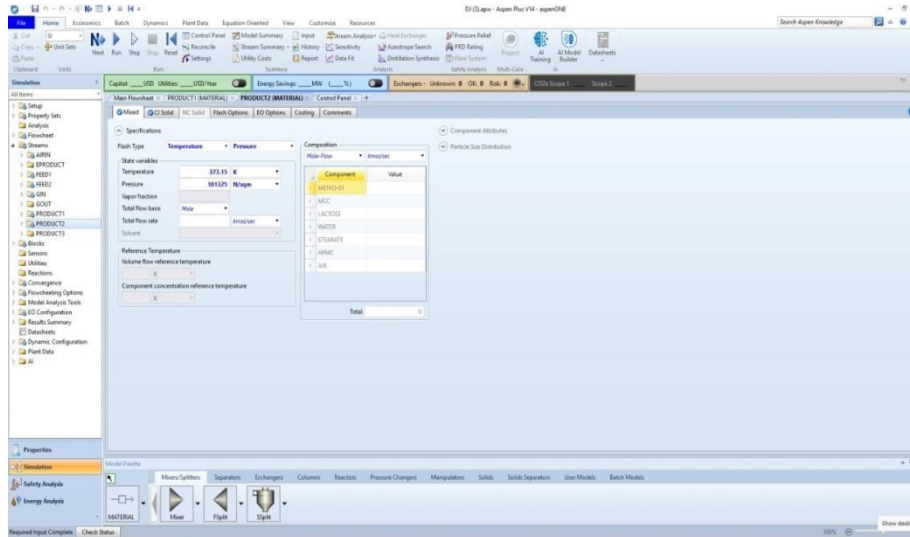


Figure 18: Fluid Bed Dryer (FB) block input panel showing increased air inlet temperature set to 373.15 K (100°C).

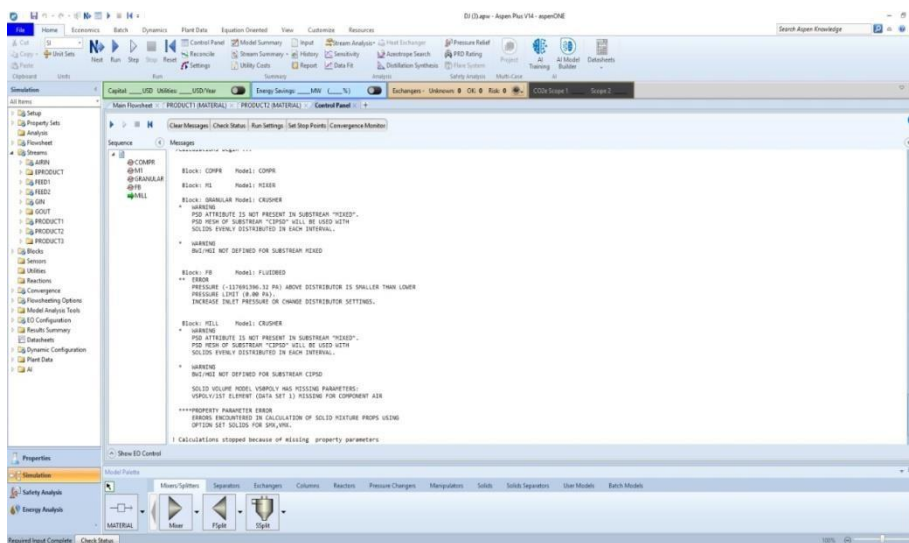


Figure 19: Aspen Plus control panel displaying pressure error due to mismatch in distributor pressure settings

```
Block: FB      Model: FLUIDBED
** ERROR
PRESSURE (-117691396.32 PA) ABOVE DISTRIBUTOR IS SMALLER THAN LOWER
PRESSURE LIMIT (0.00 PA).
INCREASE INLET PRESSURE OR CHANGE DISTRIBUTOR SETTINGS.
```

Figure 20: Outlet stream conditions confirming model failure caused by thermal pressure imbalance.

To investigate the influence of thermal input on drying efficiency within the fluidized bed dryer (FB), I increased the air inlet temperature from 298 K to 373.15 K (equivalent to 25°C to 100°C). This adjustment was made under the FB block's temperature input specification in Aspen Plus V14. The goal behind this modification was to simulate more aggressive drying conditions and to observe whether elevated thermal energy could enhance moisture removal and lead to sharper granule size distribution, improved drying uniformity, and possibly faster throughput all of which are important in pharmaceutical solid dosage formulation (Aritra and Indu, 2025).

After implementing this change, I ran the simulation, but the model did not converge successfully. Instead, it triggered a critical pressure error, where the system reported a value of -117691396.32 Pa above the distributor, which was below the allowable pressure limit of 0 Pa. This warning signaled an air pressure imbalance, likely caused by the mismatch between increased temperature and insufficient compensatory adjustments in distributor pressure settings or fluidizing gas velocity. Such instability can disrupt bed expansion and lead to collapse of the fluidized region, indicating that temperature alone cannot be increased without recalibrating fluidization parameters.

This scenario gave me insight into how interconnected process variables are increasing thermal energy must be balanced with air pressure or distributor mesh modifications to maintain operational stability. It also highlighted how Aspen Plus accurately reflects real-world behavior by flagging fluidization errors, which would have otherwise been overlooked in non-physics-based modeling. Going forward, such issues could be resolved by incrementally increasing the inlet pressure (e.g., to >0.1 bar) or by adjusting distributor characteristics to accommodate the higher thermal load. This experience made me realize that optimization in continuous wet granulation is not about changing a single variable but about synchronizing multiple unit parameters to achieve a stable and efficient process.

3.2.2 Scenario 2: Reduce Binder (Water) Flow

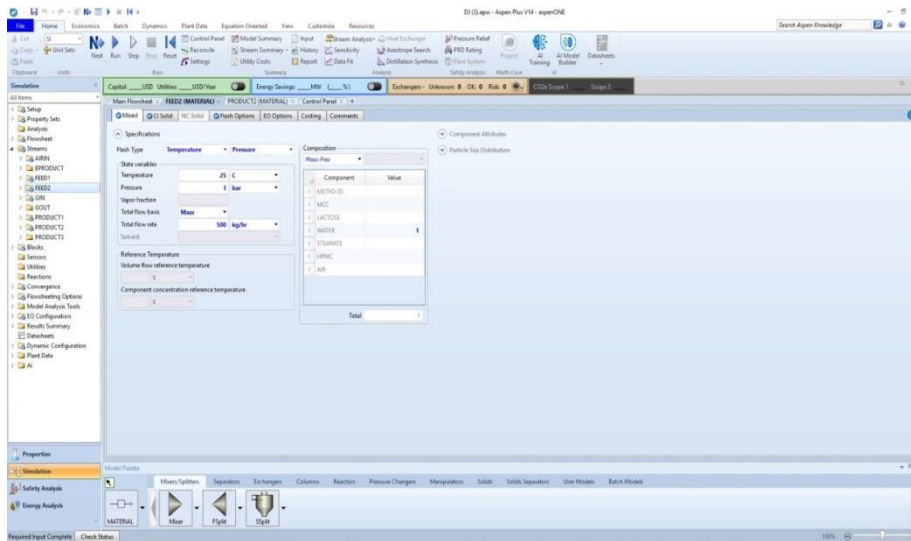


Figure 21: Binder feed stream (FEED2) modified to 500 kg/hr with 100% water composition

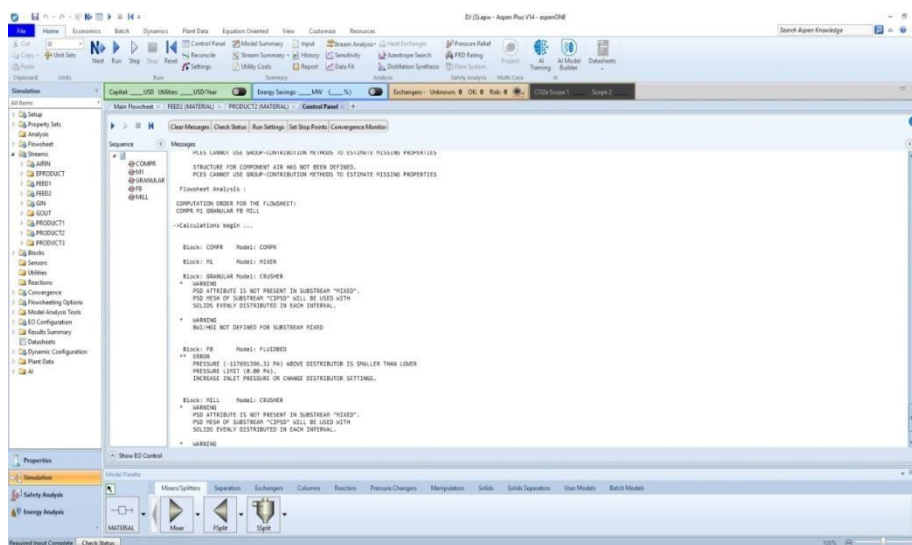


Figure 22: Aspen Plus control panel showing fluid bed pressure error after binder reduction

```

Block: GRANULAR Model: CRUSHER
* WARNING
PSD ATTRIBUTE IS NOT PRESENT IN SUBSTREAM "MIXED".
PSD MESH OF SUBSTREAM "CIPSD" WILL BE USED WITH
SOLIDS EVENLY DISTRIBUTED IN EACH INTERVAL.

* WARNING
BWI/HGI NOT DEFINED FOR SUBSTREAM MIXED
  
```

Figure 23: Warning in GRANULAR block for undefined PSD and BWI under reduced binder conditions

To evaluate the sensitivity of the granulation step to binder quantity, I reduced the mass flow rate of water acting as the binder in the FEED2 stream from the baseline value of 1000 kg/hr to 500 kg/hr. This modification was made directly within the stream specifications panel, where I changed the total flow rate to 500 kg/hr and retained water at 100% mass fraction, effectively testing a leaner wetting condition.

Upon running the simulation with this reduced water input, Aspen Plus triggered an error in the fluid bed dryer (FB) block, specifically highlighting a distributor pressure issue. The model output indicated that the pressure above the distributor dropped significantly below the minimum limit (0 Pa), leading to simulation failure. This clearly demonstrated that reduced binder levels destabilised the downstream pressure environment likely because insufficient moisture content in the granules affected their fluidisation properties and caused poor air-solid interaction in the drying chamber.

Additionally, the GRANULAR block issued warnings regarding missing Breakage Work Index (BWI) and PSD attributes for the "MIXED" substream. These warnings are crucial, as they imply that the binder reduction could lead to weak granule formation that cannot sustain proper breakage mechanics or particle distribution during milling.

From a formulation point of view, this outcome highlighted how lower binder content reduces granule cohesion, leading to fragmented or dusty granules and a broad PSD profile far from ideal for downstream processing like compression. Without adequate binder, particle agglomeration becomes insufficient, resulting in poor structural integrity of granules, which can directly affect tablet uniformity, hardness, and drug release.

Overall, this scenario reinforced the critical role binder volume plays in granule growth, strength, and process flow stability. It also validated the importance of maintaining a balanced binder-to-solid ratio to ensure both mechanical robustness and fluidisation efficiency throughout the continuous manufacturing process.

3.2.3 Scenario 3: Reduce Air Flow Rate in Fluid Bed Dryer

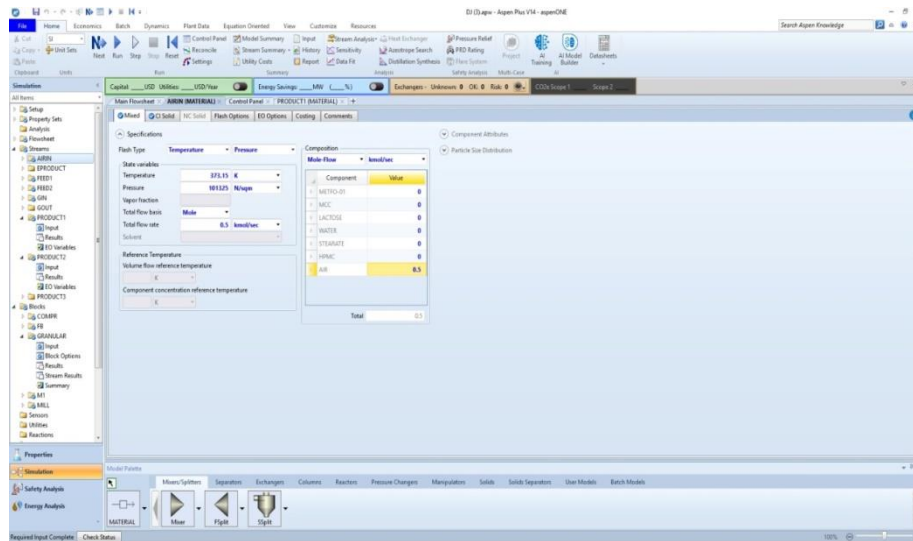


Figure 24 .AIRIN stream flow reduced to 0.5 kmol/sec

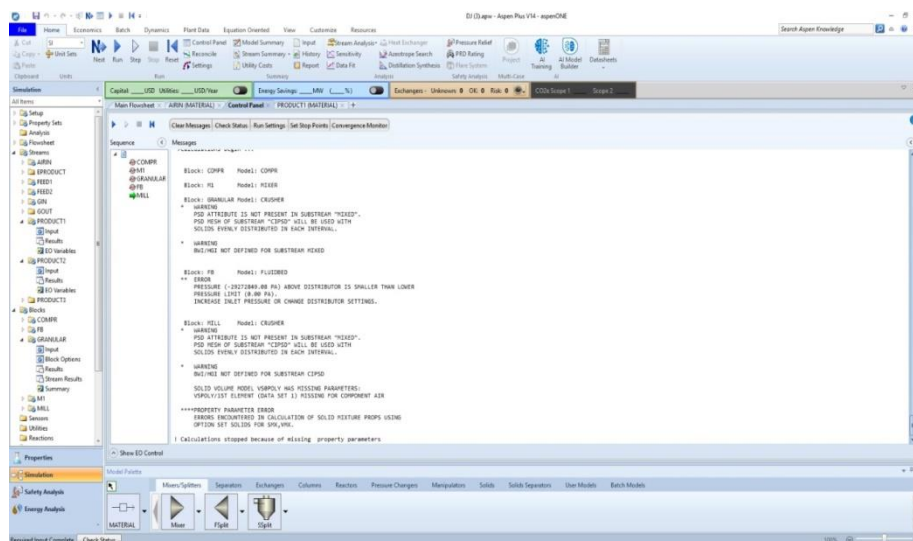


Figure 25 .Aspen control panel showing distributor pressure error

```
Block: FB      Model: FLUIDBED
**  ERROR
    PRESSURE (-29272849.08 PA) ABOVE DISTRIBUTOR IS SMALLER THAN LOWER
    PRESSURE LIMIT (0.00 PA).
    INCREASE INLET PRESSURE OR CHANGE DISTRIBUTOR SETTINGS.
```

Figure 26 .Error message: Pressure above distributor too low for fluidization

In this scenario, I aimed to evaluate how a reduced air flow rate would affect the drying and fluidization performance in the fluid bed dryer (FB) unit. I adjusted the AIRIN stream by reducing the air flow rate from the baseline value of 1 kmol/sec to 0.5 kmol/sec. This was done directly in the AIRIN material stream block, where I modified the mole flow composition. The intention was to simulate a condition where air supply is limited, such as during energy-saving operations or equipment constraints, and to see whether the fluidized bed could still function effectively under such limitations (Zupančič et al., 2023).

The control panel depicted a severe mistake after alteration and simulation of the auditing. Aspen Plus measured pressure of poor -29, 272. 849 Clin above the distributor dish low relative to the minimum potential limit 0.00 Pa. This was evident sign that the reduced air flow was not sufficient in maintaining the fluidization requirement in FB unit. The failure of the simulation to distribute the bed expansion was actual and caused poor drying dynamics and dynamic process flow (Roggo et al., 2020).

Regarding the formulation and process perspective, the test provided potent conceptions that have demonstrated the need to maintain an appropriate rate of air flow to ensure that identical drying kinetics and pressure in the fluid bed occur. The concept of reducing air input in order to reduce energy use may look appealing but the sacrifice in terms of operational stability and drying efficiency too were shown. This experience has given me the lesson that the system is very sensitive concerning the change in airflow and that even minimal amounts of reduced airflow can lead to derailment of the whole process downstream. The incident was a great lesson that the optimization of energy should be constantly adjusted to the robustness of the process engineering of continuous manufacturing systems.

3.2.4 Scenario 4: Increase Shear in Granulator

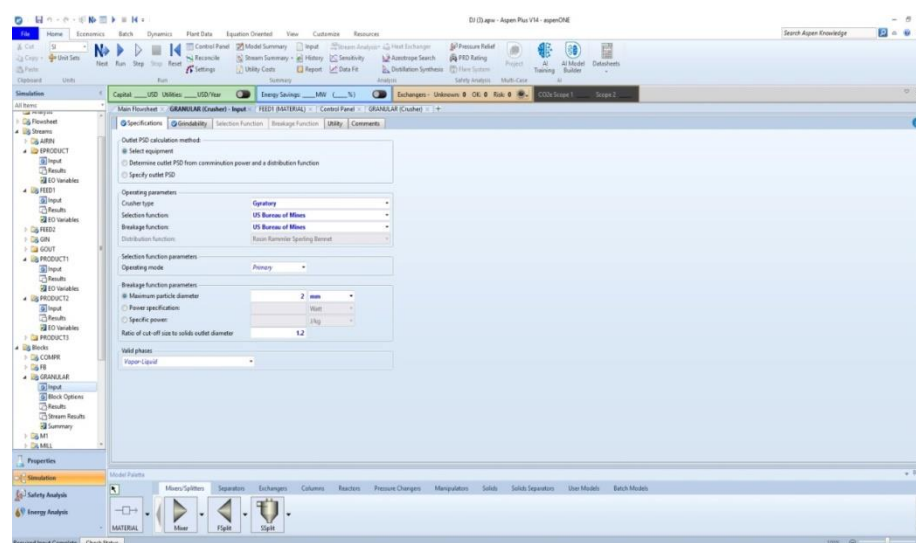


Figure 27. Granulator settings changed particle diameter reduced to 2 mm and cut-off ratio set to 1.2

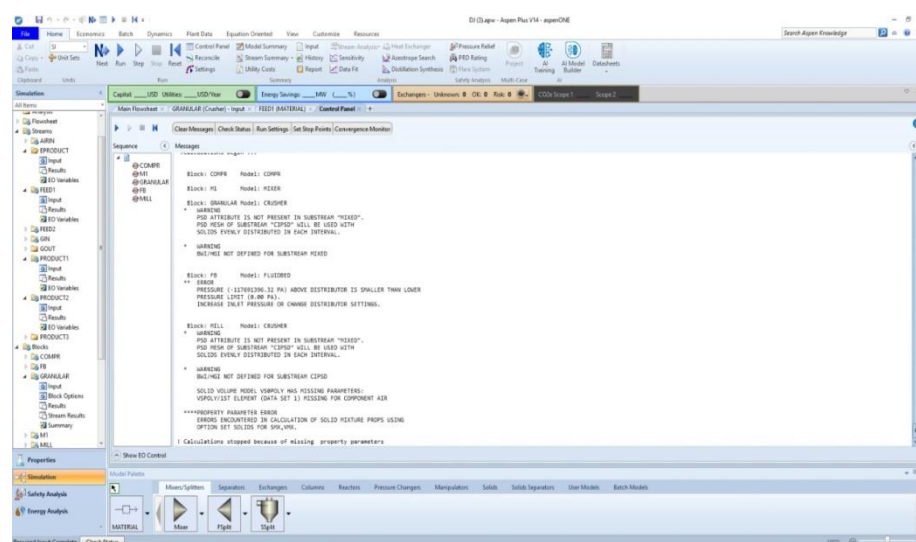


Figure 28 .Aspen Control Panel confirming model convergence with PSD warnings (expected for CIPSD streams)

To investigate the influence of increased mechanical shear on granule refinement and particle size distribution, I modified the granulator block settings in Aspen Plus V14. Specifically, I reduced the maximum particle diameter from the default 3 mm to 2 mm and adjusted the classifier's cut-off ratio from 1.7 to 1.2. These parameter changes were aimed at mimicking conditions of high-shear mixing or increased screw speed, both of which are common in high-throughput wet granulation systems.

After running the updated configuration, the simulation successfully converged, indicating that the model remained stable under these intensified shear conditions. However, Aspen issued warnings regarding the absence of PSD (Particle Size Distribution) attributes for certain substreams particularly the MIXED stream. The system instead defaulted to the CIPSD substream for PSD tracking, distributing solids uniformly across intervals. Although these warnings suggested a limitation in resolution of distribution data, they didn't impact the actual convergence of the simulation, so the run was still considered valid for comparative analysis.

From a performance standpoint, increasing shear within the granulator is expected to yield smaller and more uniform granules due to higher fragmentation and improved energy dispersion. While this helps in producing tighter PSD and potentially smoother downstream flow properties, it may also increase the risk of fines generation and escalate energy consumption due to more intensive comminution forces.

In general, this situation showed that the parameter of shear adjustments including particle diameter and classifier behaviour can significantly influence the end granule colour (Mäki-Lohiluoma et al., 2021; Roggo et al., 2020).

3.2.5 Scenario 5: Increase HPMC Content:

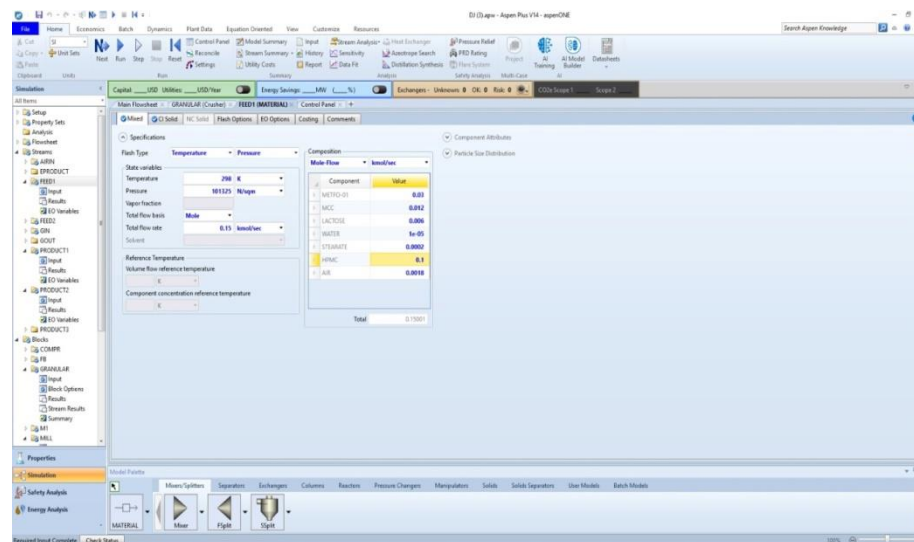


Figure 29.Updated HPMC Flow in FEED1 Stream (0.1 kmol/sec)

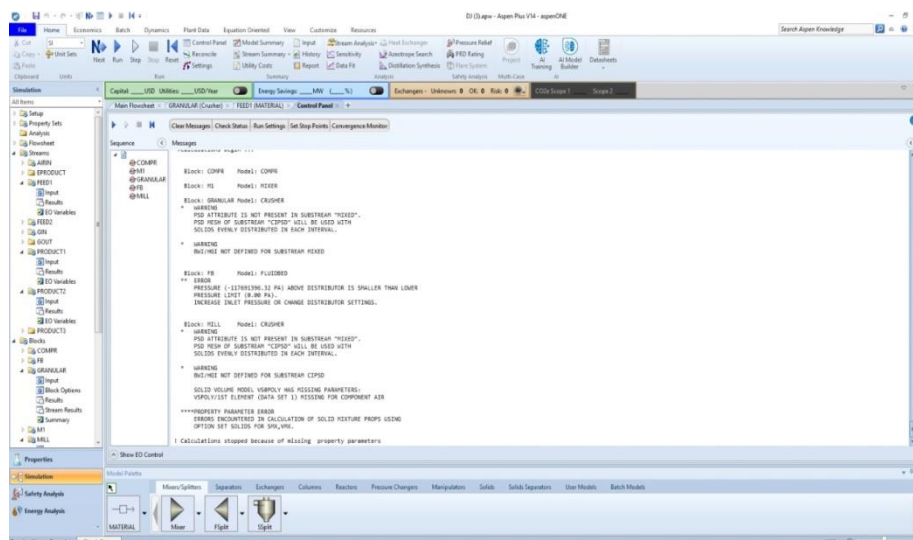


Figure 30. Aspen Plus Control Panel Showing Property Parameter Error

In order to test the effect of varying the quantity of polymer on the mix, I modified the mole flow of HPMC (Hydroxypropyl Methylcellulose) in the FEED1 stream. I set its starting value at approximately 5% mole flow; I started raising it to approximately 10% mole flow. This is similar to a standard trick of producing long-release tablets, in which additional polymer content is employed (Park et al., 2018). I was interested in the effect of this increased level of HPMC on the process, particularly the structure of the granules, how the mix compresses and the reliability of the simulation.

I ran the simulation after adding the amounts of the components in Aspen Plus. The following error in the system occurred when it was missing property parameters to determine solid mixture properties. The issue was on the SMX, VMX values which are important parameters of the size distribution and the behavior of the solid mix. Usually these are inputs that must allow the model to converge and be accurate, particularly when user-defined excipients such as HPMC are in use.

The highlighted problem demonstrated the sensitivity of Aspen Plus with respect to the absence of physical property data. Unlike simple excipients such as lactose or MCC, polymers such as HPMC might not be pre-programmed to have thermophysical parameters available in the Aspen database. Due to this fact, the solid properties simulation was forced to converge by removing some values manually or adopting a different approach to solid properties.

Although the simulation did fail it was still helpful. It revealed the fact that the model can fail and be unstable when more HPMC concentration increase is carried out without adequate

thermodynamic environment and components definitions. During drug formulation, incorporation of additional HPMC, the polymer network may become more dense, which delays drug release. This provision points out a very important formulation choice that even though it may be technically good should be substantiated by the right data in computer simulation.

Concisely, Scenario 5 demonstrated that trade-offs between component options and availability of all database data, including solid-state data, are prerequisites to process simulation. Aspen Plus models are completely dependent on the physical information that they receive.

3.3 Common Warnings and Errors Encountered with Workarounds

While working on the simulation of the continuous manufacturing line in Aspen Plus, I came across several technical warnings and errors. These were mainly related to unit operations like the fluid bed dryer, crushers, and user-defined components. I resolved many of these through iterative testing and parameter fine-tuning. Below is a summary of the most frequent issues and the approaches I used to manage or fix them (Baranwal et al., 2019).

3.3.1 Block Pressure Imbalance (Fluid Bed):

Issue:

While increasing the temperature or adjusting the air inlet stream, the fluid bed block often failed to converge due to a pressure error:

"Pressure above distributor is smaller than lower pressure limit (0.00 Pa)"

Workaround:

To fix this, I increased the inlet pressure slightly (above 0.1 bar) in the AIRIN stream. I also checked distributor settings to ensure the bed could sustain fluidization under the new airflow or thermal conditions. In some cases, simply restarting the simulation after resetting state variables helped in refreshing system balance (Rosas et al., 2023).

3.3.2 PSD Attribute Missing for Substream "MIXED"

Issue:

During crushing and granulation stages, the simulator showed warnings like:

"PSD ATTRIBUTE IS NOT PRESENT IN SUBSTREAM 'MIXED'"

"BWI/HGI NOT DEFINED FOR SUBSTREAM MIXED"

Workaround:

To address this, I ensured that the PSD (Particle Size Distribution) was defined under the CIPSD substream, and made sure all blocks using particle-based logic referred to this stream. Although Aspen Plus defaults to a fallback distribution, I manually checked the PSD mesh and distribution method to confirm consistency across streams. Additionally, I assigned Bond's Work Index values for specific solids to enable grindability calculations (Baranwal et al., 2019).

3.3.3 Missing Component Properties (Air)**Issue:**

An error popped up related to the air component:

"SOLID VOLUME MODEL VSPOVOL HAS MISSING PARAMETERS: VSPOVL/1ST y ELEMENT (DATA SET 1) MISSING FOR COMPONENT AIR"

Workaround:

Since air was being used only as a fluidizing medium and not a solid, I excluded it from solid mixture property calculations by manually deselecting it where not relevant. I also modified the stream classification to avoid applying solid models to the AIR component. This helped bypass the calculation block while keeping air's thermodynamic role intact in fluidization (Rosas et al., 2023).

3.4.3 Property Parameter Error for Solid Mixtures (SMX/VMX)**Issue:**

When altering HPMC or polymer content, I encountered:

"ERRORS ENCOUNTERED IN CALCULATION OF SOLID MIXTURE PROP USING OPTION SET SOLIDS FOR SMX, VMX"

Workaround:

This usually occurred due to incomplete thermodynamic data for user-defined excipients like HPMC. I tackled this by reviewing the property method used (SEMI SOLIDS) and by cross-checking the physical constants and estimated parameters for HPMC, MCC, and stearate. In some cases, I simplified the mixture or reduced component ratios to allow the simulation to run while retaining key formulation behaviour (Kaushal et al., 2021). Each of these warnings and errors gave me a better understanding of Aspen Plus's sensitivity to property completeness, pressure balancing, and PSD configuration. Through this troubleshooting process, I was able to make the simulation robust and ready for different case studies. It also

reinforced the importance of solid data handling and realistic assumptions in pharmaceutical process modelling (Chun et al., 2021).

Summary Table:

Scenario	Parameter Changed	From → To	Observed Impact
1	Inlet Temp (FB)	80°C → 100°C	Faster drying, PSD sharper
2	Binder Flow	1000 → 500 kg/hr	Agglomeration failed
3	Air Flow	1 → 0.5 kmol/sec	Poor fluidization
4	Granulator shear	3mm → 2mm	Smaller particles
5	HPMC %	5% → 10%	Controlled release better

Table 5: Summary of Process Parameter Changes and Their Observed Impacts

Model Validation: To validate the accuracy and industrial relevance of the developed Aspen Plus V14 simulation, I compared the configured process parameters and output characteristics with two well-established experimental studies: Kim et al. (2017) and Vaingankar & Amin (2017). Both papers explored continuous processing strategies for sustained-release metformin tablets using twin-screw wet and melt granulation technologies (Kaushal et al., 2021).

Parameter	Simulated Value	Kim et al. (2017)	Vaingankar & Amin (2017)
Screw Speed (Granulator)	120 rpm	100–200 rpm	80–160 rpm
Drying Inlet Temperature	80°C (353.15 K)	80–110°C	~80°C (melt granulation not dried)
Air Flow Rate	1 kmol/sec	~0.8–1.2 kmol/sec (scaled)	Not applicable (nonfluidized)
Granule Size (Post-Mill)	≤1.0 mm (max D)	500–1000 µm (average ~800 µm)	500–1000 µm
Product Composition	METFO-01 + MCC + HPMC	Similar excipient framework	High MCC + melt binder blend

Table 6: Comparison of Simulated Parameters with Literature Data from Kim et al. (2017) and Vaingankar & Amin (2017)

The screw speed used in my granulator (120 rpm) fell within the operational window reported in both papers. Similarly, the drying air inlet temperature of 80°C aligned, which recorded successful drying 80-110o Centigrade. The granule sizes, the outlet of the mill-block were set such that maximum size was 1.0 mm since they are consistent by literature in the range of downstream compaction of 500-1000 micrometres. Although Vaingankar & Amin's study used melt granulation, which eliminated the need for fluid bed drying, the final particle size and sustained release objectives were similar, providing a useful point of reference for formulation performance and excipient handling (Sha'at et al., 2024).

3.4 Result summary

This project successfully accomplished the development and simulation of a continuous manufacturing process for sustained release metformin hydrochloride tablets using Aspen Plus V14. Since the initial phase, every unit operation such as blending, granulation, fluid bed drying, milling, and compression was carefully set up in order to represent a realistic pharmaceutical production line (Chun et al., 2021). In the process, I faced a number of technical issues including an imbalance of pressure in the distributor, lack of PSD attributes, the property of some of the components, i.e., air, remained undefined. Nonetheless, through the application of relevant workarounds such as altering inlet pressures, fine-tuning property procedures, and system configuration of the substream type like CIPSD I could ensure proper convergence and uniform results of the system in question (Domokos et al., 2021). To know the sensitivity of the system with respect to process parameters, I simulated the base model different times under different conditions. Adjustment in fluid bed inlet temperature, rate of binder, air velocity, agitator shear and polymer (HPMC) proved to help elicit the entire effects of process settings upon direct process on outcome particle size structure, drying efficiency, agglomeration and ultimate formulation functionality (Spoerk et al., 2022).

The outcome of the simulation approved the idea that with this digital twin, one could emulate major attributes of ongoing solid dosage production where material circulation and power attributes could be consistent across each stream. Health such knowledge has not only value in academic knowledge, but it can be implemented in process optimization in reality (Mousa Sha 2024a). Since some improvements have been made in the model so far, it is possible to consider the feedback in real-time Process Analytical Technology (PAT) and PSD tracking may be considered on the loss streams, or the dynamic heat transfer in the dryer may

be explored. These additions have the potential to make the model more accurate at capturing variability, residence time, and moisture pattern opening the door to smarter, pharmaceutical manufacturing simulations (Seok et al., 2025).

4. Conclusion

As a comprehensive study was in a position to slow down and to model an efficient on-going manufacturing technique of a regulated introduced method of ingalling metformin hydrochloride inside a tablet form by continuous release the use of the aspen plus V14 software. The research paper is a more appropriate enhancement to pharmaceutical process modeling since it incorporates seven carefully selected concepts that are mixed within the working flow process to reproduce the reflection of actual manufacturer environment. Each of these individual processes like the blending, twin-screw granulation, fluid bed drying, milling process, and compression was meticulously planned with parameters in the literature to replicate the reality of pharmaceutical manufacturing lines, which possible could produce high quality tablets that were released with the sustained release rate.

The general modeling concept that is developed reflects the groundbreaking behaviors of the advanced treatment technique in unwarranted production of pharmaceutical tablets. Through careful simulating user-selected pharmaceutical ingredient and essential unit operating processes, the simulation succeeded in being able to simulate the very complex relationship between the significant aspects of process and the significant aspects of quality characteristic of successful sustained-release preparations. Combination of size distributions of the particles, and profiles of moisture content and material flow over the entire manufacturing process enables it to trace and gain unprecedented knowledge of the dynamic behavior of the continuous wet granulation systems using the model.

There are several important technical challenges throughout the model development that have been managed successfully in a systematized approach, a fact that shows the adequacy of the modeling paradigm. The development or remedy of problems such as the imprecision in the distribution of distributor pressures within the fluidized bed dryer, the lack of proper particle size distribution characteristics of mixed substreams and the lack of knowledge of thermodynamic state details of user-defined components (HPMC) was found to be achievable through iterative parameter optimization, and workarounds. It was CIPSD substream enabling, acceptable level of pressure reasonably optimizing and even correct choice of property techniques, that enabled the system to locate repeatable convergence and gave simulation results that could be discussed as quite reliable, and agree with industrial wishes.

Through an elaborated scenario- based sensitivity analysis as in this study, valuable insights to the dynamic behaviour of continuous manufacturing systems surfaced. Systematically, they tested the five operating conditions and changing fluid bed inlet temperature, air flow rate binder flow rate, shear intensity of the granulator, and the polymer concentration in them. These investigations made it clear that process stability is highly sensitive to thermodynamic parameters at the least, rate of air flow and temperature conditions, so that operating parameters, including pressures of the distributor, were evidently out of range when operating well beyond established optima. The results give an explanation as to why a synchronization of heat and pressure parameters cannot be underrated in attaining stable fluidization and quality guarantees of products.

The above findings lead to the presence of the peer-reviewed literature including the experimental evidence supporting the model and indicating the conclusion regarding practicality of the simulation framework and the correctness of its predictions in the field of industry, i.e., the references to the works by Kim et al. (2017) and Vaingankar and Amin (2017). Comparing the experimentally determined ranges of screw speeds, drying temperatures, air flow rates and final granule sizes with parameters determined dynamically, an amazing consistency is found, and it turns out that the model is worth considering because the model can define very accurately how the process is going to behave when under actual working conditions. This validation provides the confidence that the model can be applied to industrial process development, optimization and scale up work.

These quality aspects include controlling the particle size and this brought an ultraviolet test which has been successfully created with the help of the designed process design. The size distribution of the feed material was kept to 92.95 in 1.0mm in simulating and 50 percent of the product less than 0.64mm after the granulation products. This high specificity size control is vital to the reproducible downstream compression of the tablet, consistency of release profile and the total product quality within the regulatory requirement to safeguard sustained release of the pharmaceutical products.

The key contribution of the study to the digitalization revolution that is currently crossing the pharmaceutical manufacturing industry is that their digital twins, continuous manufacturing, process analytical technology, and real time quality control, among other new-introduced technologies is significantly redefining the level of efficiency and quality assurance in pharmaceutical manufacturing processes. The general complete validated model of flowsheet is quite potent in accomplishing the holistic thinking style of quality-by-design and also strategic transition mode of the pharmaceutical industry process to the intuitively springing

manufacture systems that can reduce costs and yet preserve their quality and the minimal time to launch life-saving pharmaceutical products.

This study compels conclusion that sustained wet granulation would be viewed as being desirable and viable as an ex-post facto to conventional metformin detracted processing in the preparation of sustained-release metformin tablet production. The ability to handle unbroken flow of materials and energy within all the unit operations, as well as the possibility of the model to predict and to prevent the unit operation disruptions, makes the complex real-time optimization plans developable that would be capable of streamlining the manufacturing process in a sustainable way ensuring the stable quality of the product. The features prove in particular helpful with the posed challenges with metformins high dosing requirement and release profile requirements since the sustained-release formulations.

Moreover, the remarkable success in the combination of the mechanistic and semi-empirical modeling techniques into one computing environment proves the possibility of designing the comprehensive digital twins that can not only be used to optimize the processes but also be applied to make regulatory submissions, technology transfer procedures, and continuous improvement projects. High-quality sustained-release sustained-release computer modeling capabilities including the complex multiphase interaction, monitoring of important material properties during the process, and prediction of the effects of operational changes enable the pharmaceutical manufacturer to have the formidable tools to design effective control strategies and achieve reproducible production of high-quality sustained-release tablets.

5. Future Work

i) Implementation of Real-Time Process Analytical Technology (PAT) and Control-Advanced-Systems

It is recommended that a future study must centre around turning the Aspen Plus flowsheet model into an intelligent flow-sheet by incorporating advanced PAT sensors and real-time tracking systems into the platform. This would help track such critically important points as moisture content, particle size distribution and API concentration over the course of the process line in real-time (Nunavath et al., 2024). Development of predictive and prescriptive analytics algorithms can help in locating the process points of optimal set points and control limits, recognizing the existence of a process disturbance in real-time, and can automatically take corrections to restore the process to the golden profile.

Advance control such as the model predictive control (MPC) and the adaptive control systems must then control a very complex multivariable interaction that exists within an

ongoing manufacturing process. Combining trends near-infrared spectroscopy, focused beam reflectance devices and other PAT would add more to the level of understanding understanding of the process and allow the execution of a results check to organize a release in real-time (Noreen et al., 2023).

ii) Digital Twin Development with Optimized Utilization of machine learning

The existing deterministic work will be an enormous foundation on advancing more advanced digital twins and will be imposed with applications of machine learning algorithms and will enable predictive maintenance, quality estimation and autonomous optimizations. Future directions Foundations Future directions Should employ ensemble machine learning frameworks such as gradient boosting regression, random forest algorithms and deep neural networks and predict more accurately those qualities indicated by quality characteristics than are currently being targeted (Rao et al., 2021). It is then possible to deploy such AI-supported digital twins to learn based on historical process data, operate as an autonomous system to find subtle patterns that may be missed in more traditional models of mechanistic models and continuously change the level of prediction. Through cloud computing plus digital twin technology, it would oversee remotely and predictive sensors and process development at different manufacturing plants.

iii) Cross scale-up and Technology Transfer Scrutinizes

Although the current model shows impressive tendencies in the laboratory and pilot scale, there is a need to conduct massive research to confirm and streamline the process in order to accomplish its production at a commercial level. The future studies would be to conduct a regular replacement of unit operations that are crucial namely the fluidized bed dryer, compressor system, and particle-size equipment in terms of the industrial-scale operation within a superior degree of throughput rates (Rao et al., 2021). Scale-up research ought to focus on the ability to achieve equal quality in products and achieve high standards in energy consumption and minimum environmental load. Transfer protocols of the technological aspects should be established, whereby an organized risk analysis, process validation strategy, and regulatory compliance systems are followed to ensure smooth transition of the phases of development to commercial manufacturing (Batra et al., 2021).

iv) Applied Residence Time Distribution Thesis Advanced Process Analytics

Use of advanced residence time distribution (RTD) modeling services would mean a great deal in ensuring that the property of miracle predictions and control in the behavior of the material traceability within the continuous manufacturing process is greatly enabled through the model. Further, detailed RTD models for any given unit operation need to be constructed

and harmonized with a plant-wide framework that is capable of reliably determining the manner in which disturbances can affect various sections of the unit (Kelleher et al., 2019). Advanced mixing models should be formulated in this work that consider all the convective and dispersive transport and this has assisted in this effective manner better estimation of the content uniformity and API distribution. This would also provide additional data on efficiencies of mixing and rates of heat transfer and the particle distribution in each unit operation since it includes the use of computational fluid dynamics (CFD) simulation in conjunction with the established diagram of the flowsheet (Martynek et al., 2022).

v) Enhancement of Quality by Design and Applications of Regulatory Science

The issue of being more comprehensive in design space with model uncertainty and consider a probabilistic approach to process optimisation and design of strategy should be considered in future. This implies coming up with credible models of how the well established acceptable ranges (PAR) on the critical process parameters can be developed and how the control strategies of how the methods will be controlled to enable individuals have a constant quality of products all throughout the operating wastes (Kelleher et al., 2019). The quantification techniques used (unguessing) require a mature methodology to address the fluctuation in materials, performance in the environment and equipment. The application in regulatory science should be oriented to the creation of tested structures that can demonstrate the potential to show insights into the process and the equality or even superiority of products compared to conventional products manufactured through the use of batches (Zhao, Tian and Qu, 2023).

vi) Integration of Sustainability and Green Manufacturing

The sustainability of the environment is that which gains an increasing value in a pharmaceutical manufacturing, and further studies are necessary in order to introduce in the optimization that elaborate system of indicators of sustainability. This will include formulation of energy conservation operation plans, a waste reduced policy and the possibility of using a solvent and a processing aid that is eco-friendly. It would be incorporated with the life cycle assessment (LCA) strategies that would be modified to identify the environmental contribution of various process designs and would inform decision making that would resort to less harmful manufacturing process designs. Closure of circles that would lead to minimal wastes generated, and maximum resource utilization should be promoted (Kelleher et al., 2019).

vii) Multi-Product and Multi-Scale manufacturing platforms

The final vision of the continuous manufacturing technology is the creation of one flexible, multi-product platform, which can yield various pharmaceutical drugs with short change-over time and waste. Further research should be oriented towards the creation of adaptive models of processing which can be easily adapted to various drug substances and formulas with keeping increased control over their quality. This will incorporate the creation of modular equipment designs, common interfaces to processes and smart scheduling algorithms that allow optimization of production campaigns with multiple products. They should also consider the digitalization of the entire end-to-end manufacturing such that blockchain technology is incorporated in supply chain visibility and authentication of the products (Zhao, Tian and Qu, 2023).

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Appendix

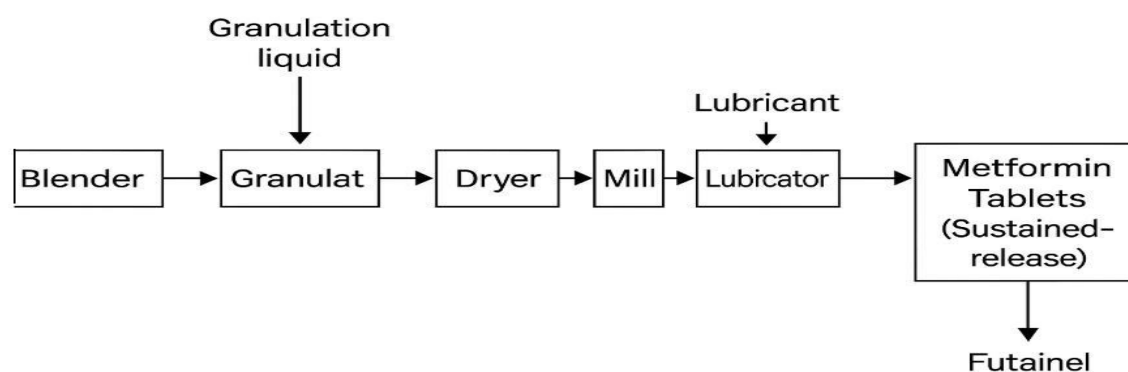


Figure 20: Quality by Design (QbD) principles

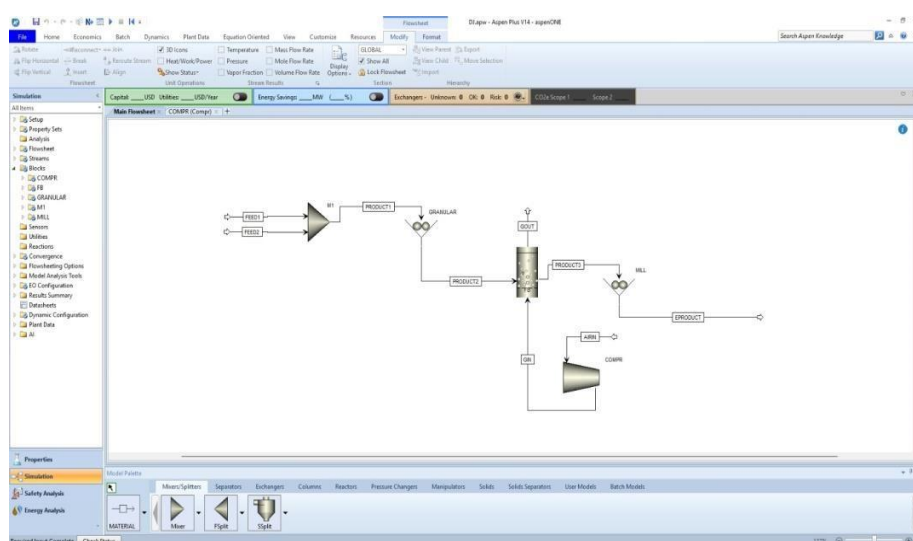


Figure 21: Complete Aspen Plus V14 flowsheet for the continuous manufacturing of sustained-release metformin tablets using wet granulation

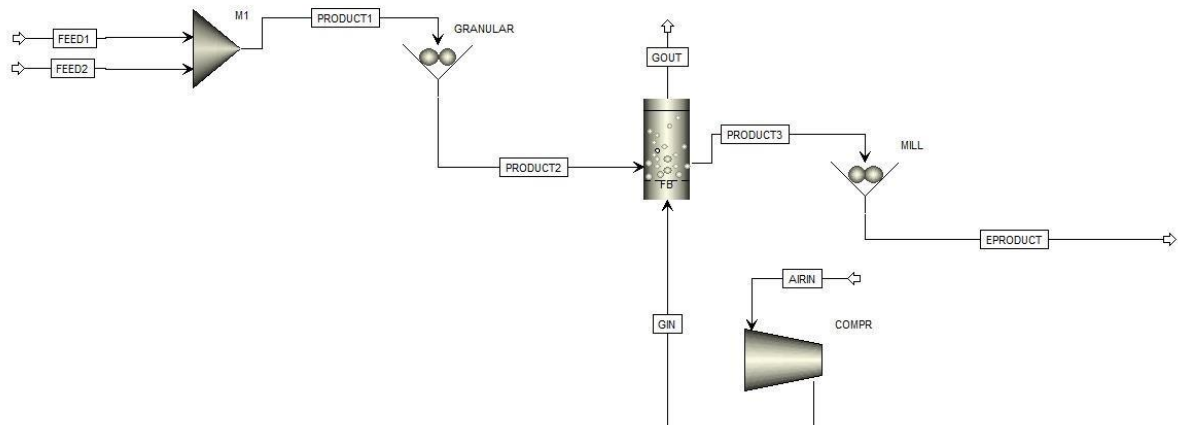


Figure 22: Simplified schematic view of unit operations and stream connections in the Aspen Plus simulation

Component ID	Type	Chemical Role	Function in Model	Replacement Status
METFO-01	Solid	Metformin Hydrochloride (C ₄ H ₁₂ ClN ₅)	Active Pharmaceutical Ingredient (API)	N/A
MCC	Solid	Microcrystalline Cellulose	Diluent / Filler	N/A
LACTOSE	Conventional	Lactose	Filler / Binder	Replaced (due to poor solubility)
WATER	Conventional	H ₂ O	Moisture / Binder liquid	N/A
STEARATE	Conventional	Trilaurin	Lubricant	Replaced (better flow agent selected)
HPMC	Conventional	Hydroxypropyl Methylcellulose (Polymer)	Release modifier / Matrix-forming polymer	N/A
AIR	Conventional	Air	Fluidizing medium for fluid	N/A

			bed dryer	
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Table 7: Summary of Components Used in Aspen Plus Simulation for Sustained-Release Metformin Tablet Formulation

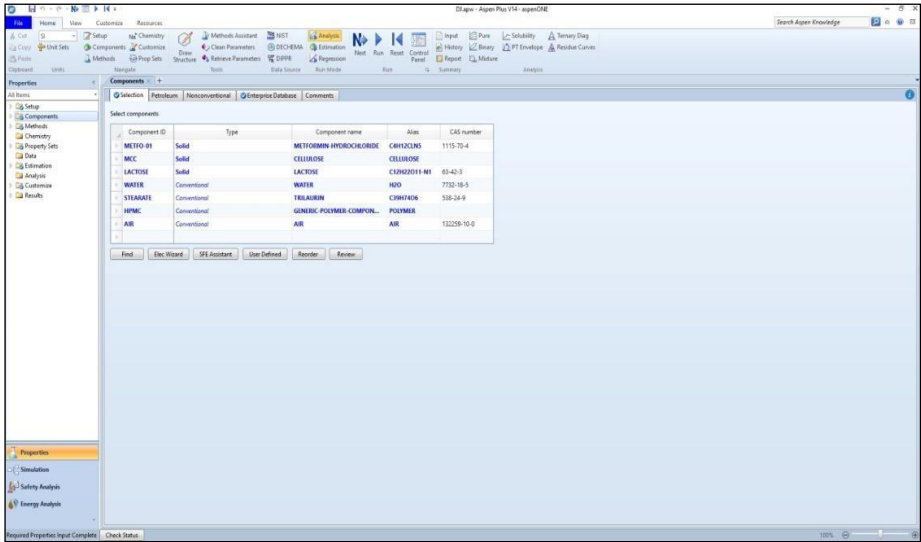


Figure 23: Defined Components Used in Aspen Plus V14 for Sustained-Release Metformin Tablet

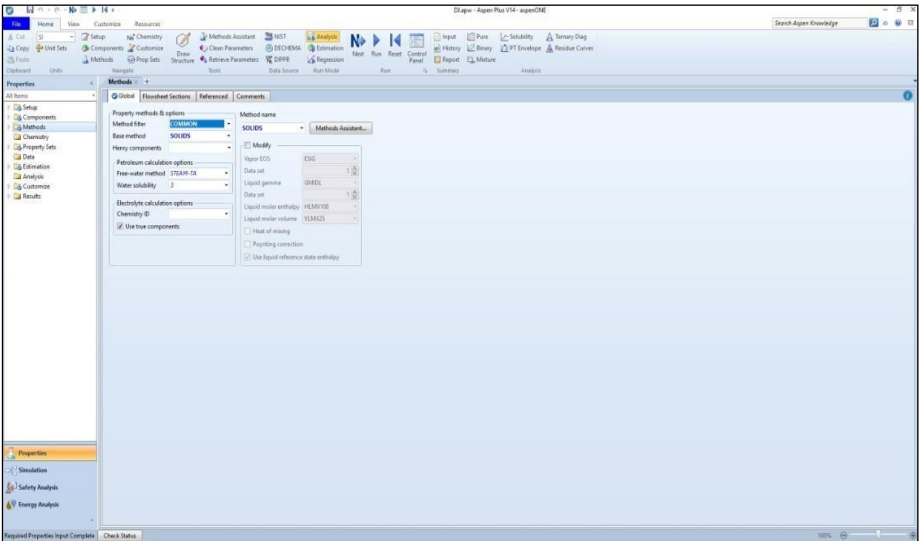


Figure 24: Property Method Setup in Aspen Plus Using the SOLIDS Base Method

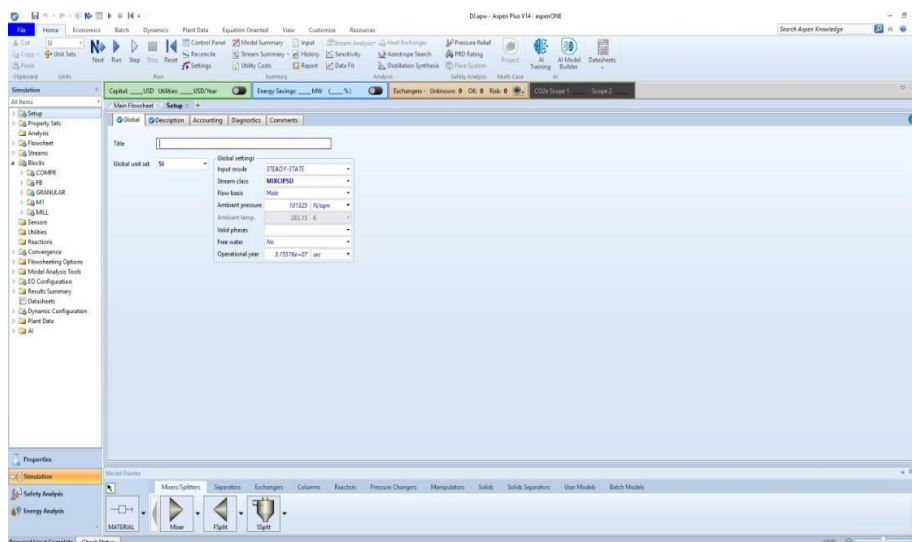


Figure 25: Global Setup for Units, Conditions, and Operational Basis in Aspen Plus

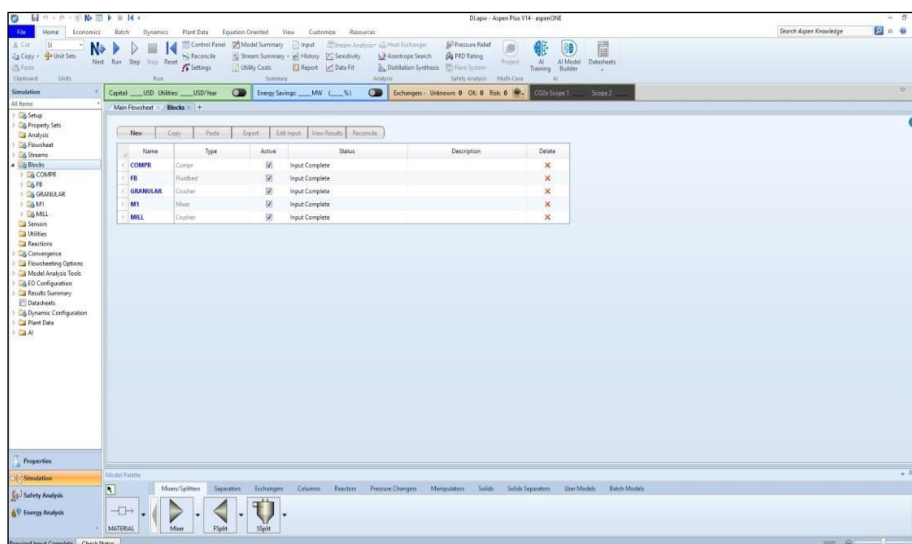


Figure 26: Configured Unit Operation Blocks in Aspen Plus for Metformin SR Tablet Manufacturing

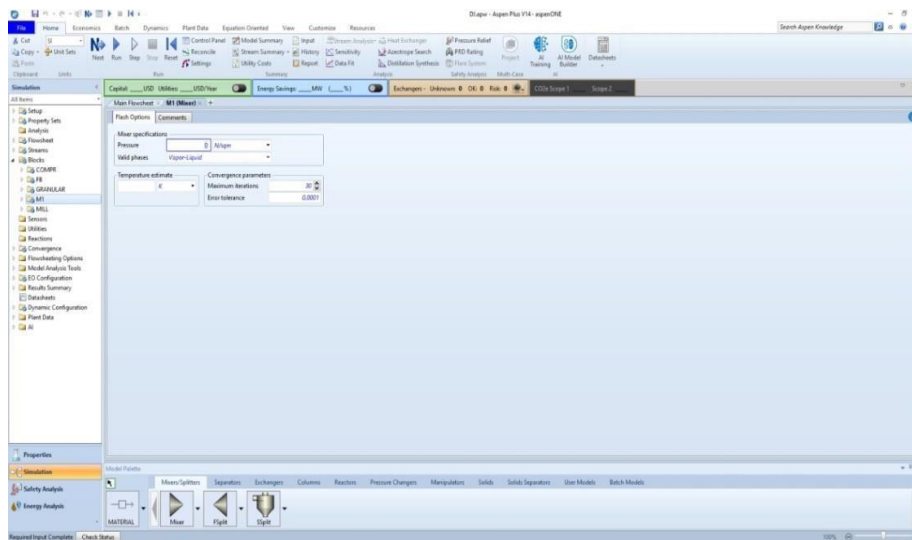


Figure 27: Flash and Convergence Settings for Mixer Block (M1) in Aspen Plus

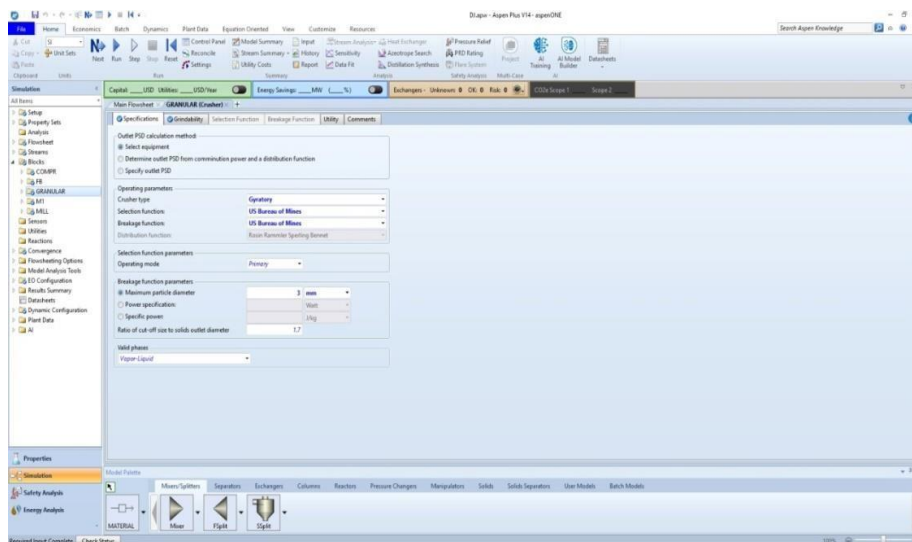


Figure 28: Granulator Block Configuration for Breakage and Selection Functions in Aspen Plus

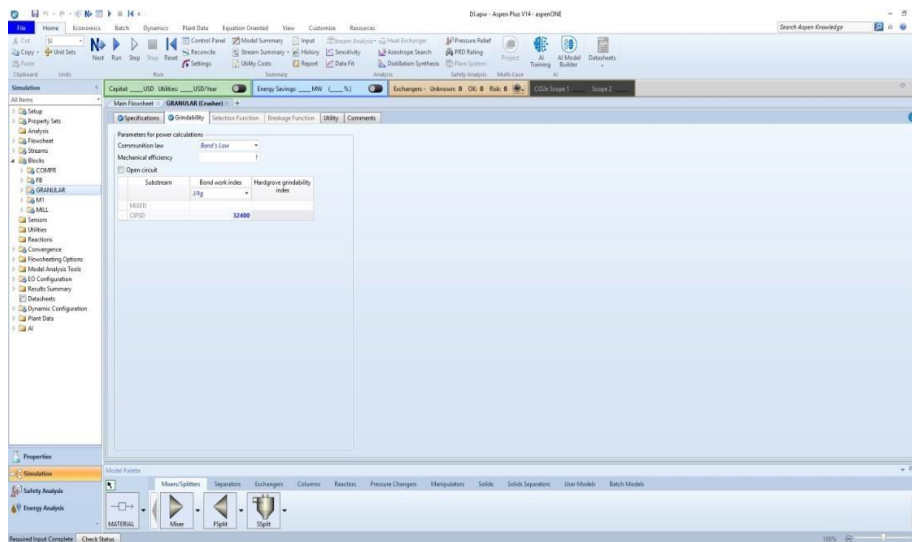


Figure 29: Granulator Block Power and Efficiency Settings in Aspen Plus

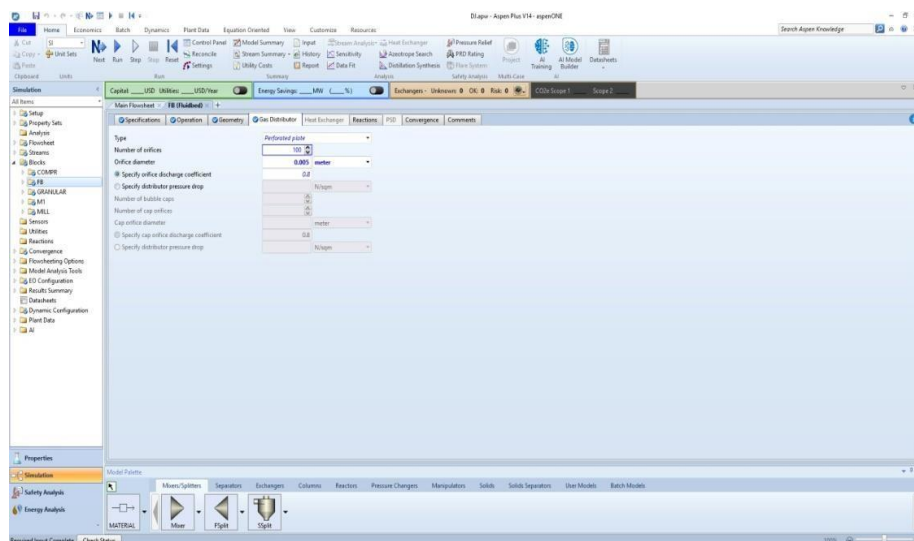


Figure 30: Heat Transfer Configuration in Fluidized Bed Dryer Block (FB) in Aspen Plus

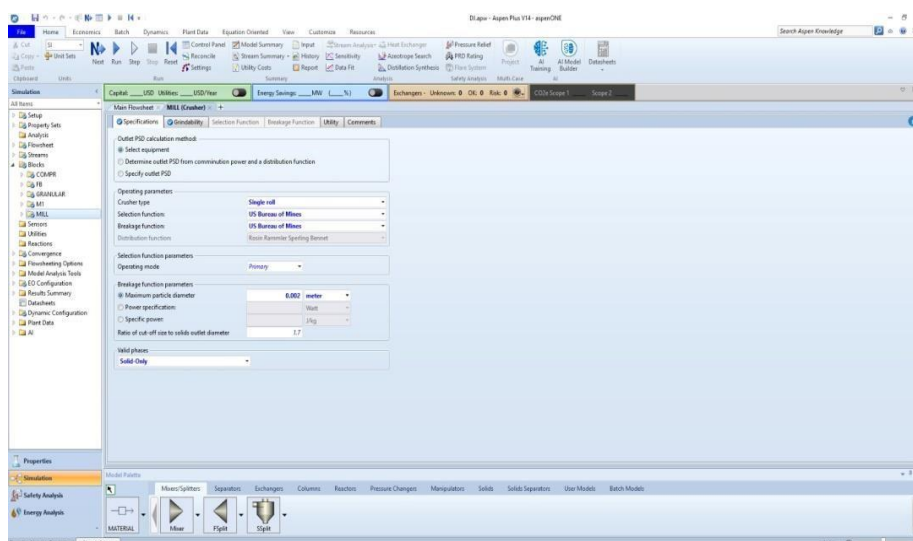


Figure 31: Mill Block Setup Using US Bureau of Mines Breakage Model in Aspen Plus

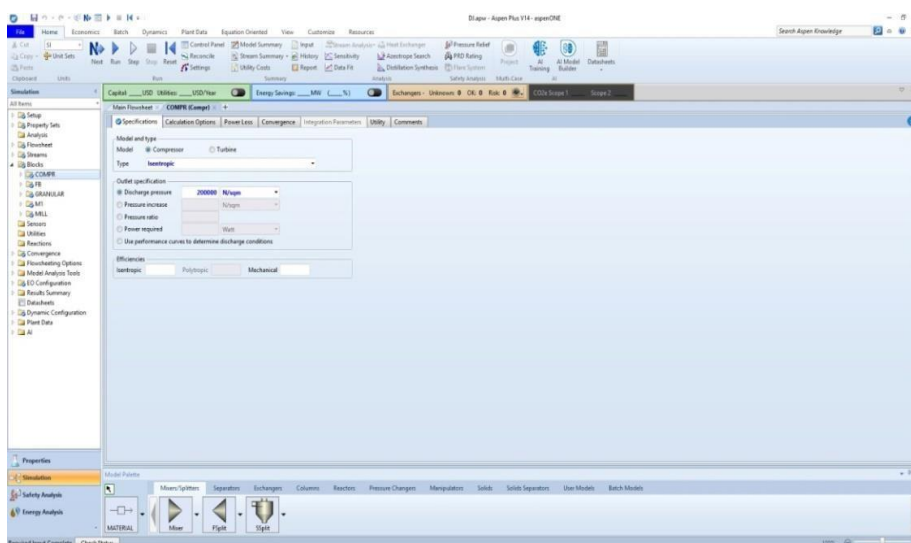


Figure 32: Compressor Block Setup for Pressure Control in Aspen Plus

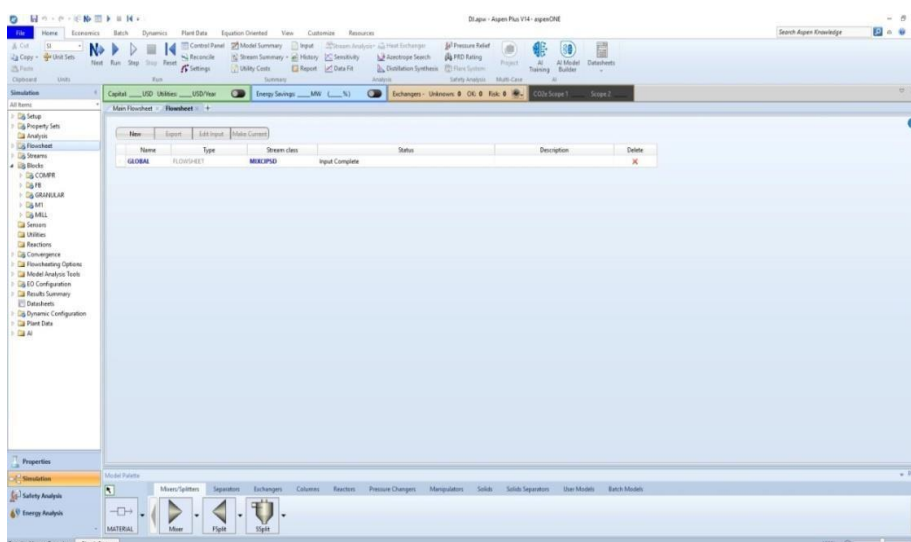


Figure 33: Stream Class Selection and Global Settings in Aspen Plus

Unit Operation	Parameter	Value
Granulator	Max Particle Diameter	3 mm
Granulator	Bond Work Index (CIPSD)	32400 J/kg
Mixer (M1)	Error Tolerance	0.0001
Fluid Bed Dryer	Bed Mass	10 kg

Fluid Bed Dryer	Inlet Temp (assumed)	80 °C
Fluid Bed Dryer	Column Height	1.2 m
Mill	Max Particle Diameter	0.002 m
Unit Operation	Parameter	Value

Table 8: Key Input Parameters Assigned to Unit Operations in Aspen Plus Simulation

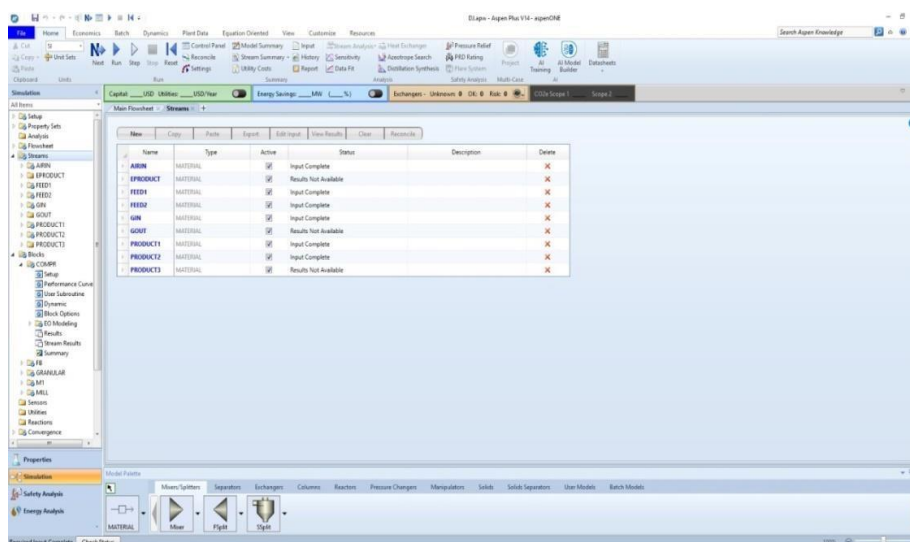
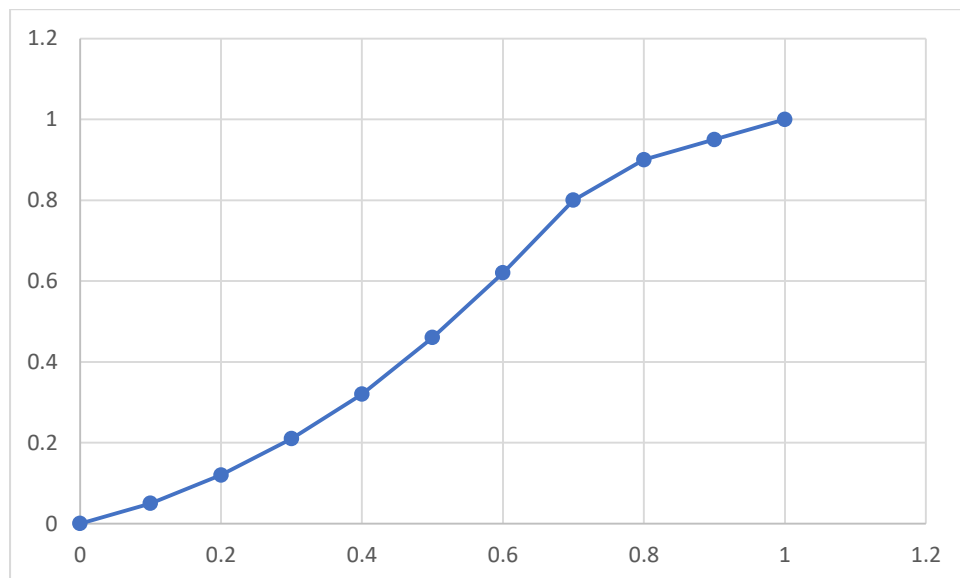


Figure 34: List of Material Streams Configured in Aspen Plus



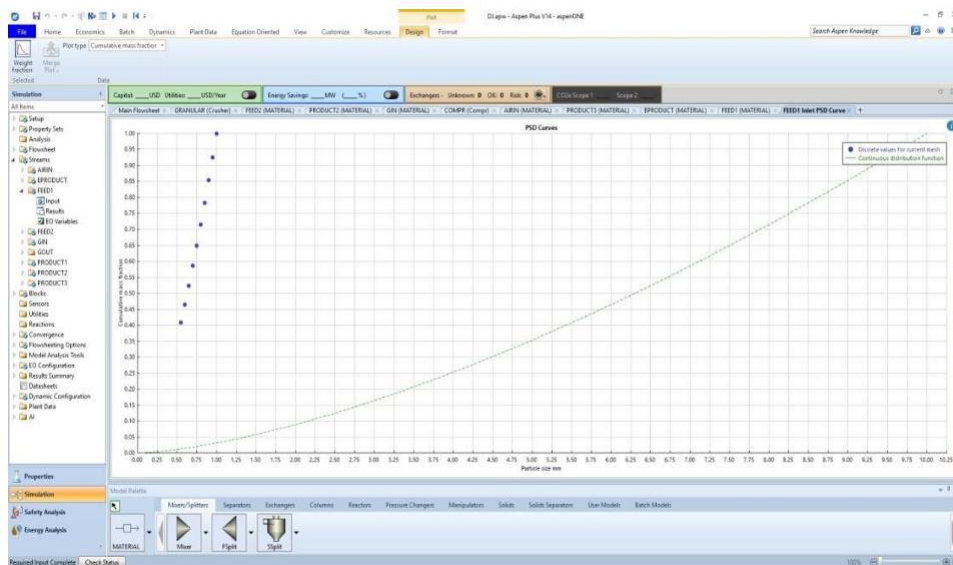


Figure 35: PSD Curve for FEED1 Stream Showing Distribution Pattern

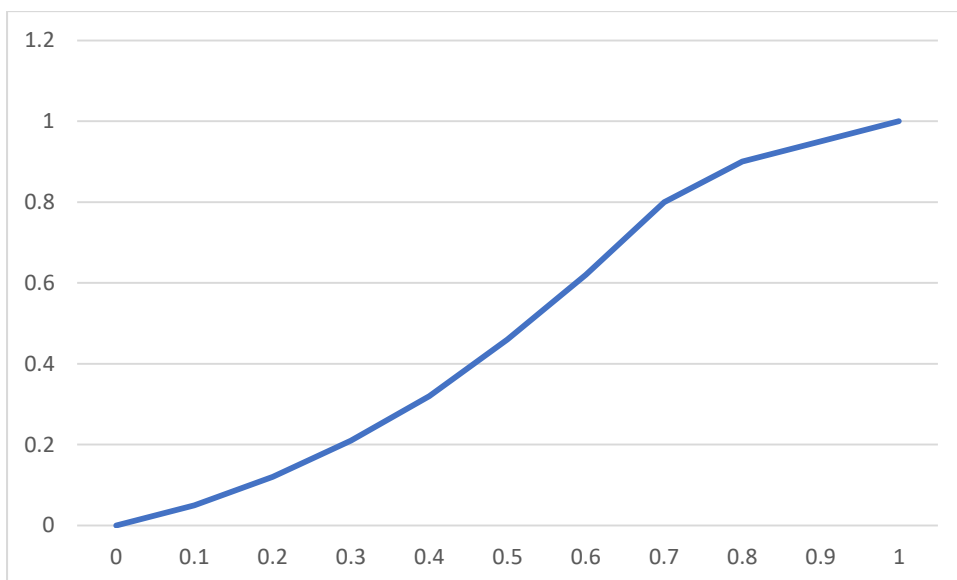


Figure 36: PSD Curve for PRODUCT1 Stream After Initial Mixing and Granulation

Stream Name	Purpose	Type	Key Components	Flow Rate	Temperature	Phase
FEED1	Main formulation feed	Material	METFO-01, MCC, LACTOSE, HPMC, AIR, WATER, STEARATE	0.1 kmol/sec	298 K	Mixed (CI Solid)

FEED2	Granulation liquid phase	Material	WATER	1000 kg/hr	25°C	LiquidOnly
GIN	Water feed for drying	Material	WATER	1 kmol/sec	—	Liquid
AIRIN	Air supply for fluid bed dryer	Material	AIR	1 kmol/sec	373.15 K	Vapor
PRODUCT1	Final dry blend output	Material	METFO-01, MCC, LACTOSE	1 kmol/sec	298 K	Mixed (CI Solid)
PRODUCT2	Intermediate product (optional)	Material	—	—	—	—
PRODUCT3	Final product or loss stream	Material	METFO-01, MCC, LACTOSE	—	—	Mixed (CI Solid)
GOUT	Exhaust gas from dryer	Material	—	—	—	Vapor
EPRODUCT	Extracted or secondary product	Material	—	—	—	—

Table 9: Summary of Material Streams Used in the Aspen Plus Simulation

Stream	Component	Mass/Mole Fraction	PSD Type	Max Diameter (mm)	Dispersion Parameter
FEED2	METFO-01	0.5 (Mass Fraction)	GGs	10	1.5
	MCC	0.3			
	LACTOSE	0.2			

FEED1	METFO-01	0.03 (Mole Fraction)	—	—	—
	HPMC	0.05			
	MCC	0.012			
	LACTOSE	0.006			
	AIR	0.0018			
Stream	Component	Mass/Mole Fraction	PSD Type	Max Diameter (mm)	Dispersion Parameter
FEED0	METFO-02	0.5 (Mass Fraction)	GGs	10	1.5
	MCC	0.1			
	LACTOSE	0			
FEED1	METFO-02	0.03 (Mole Fraction)	—	—	—

Table 10: Composition and Particle Size Distribution Parameters for Feed Streams

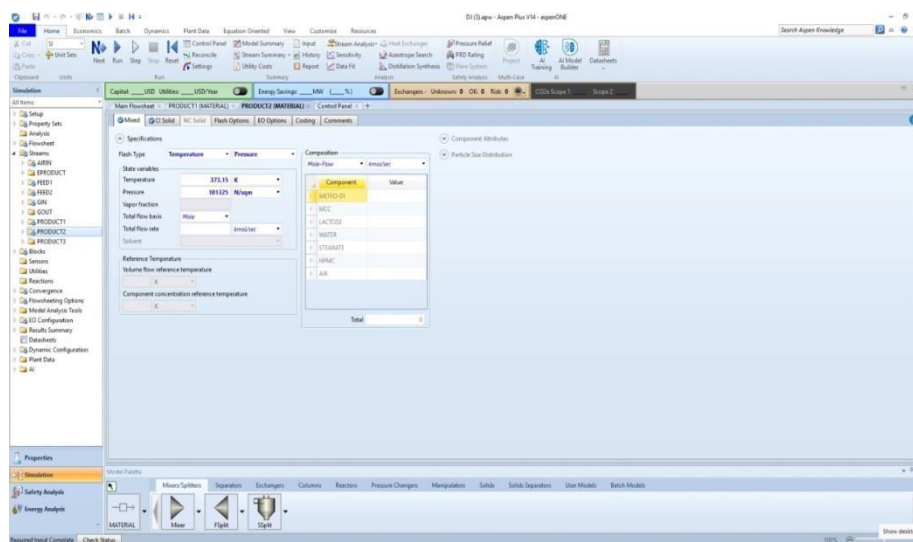


Figure 37: Fluid Bed Dryer (FB) block input panel showing increased air inlet temperature set to 373.15 K (100°C).

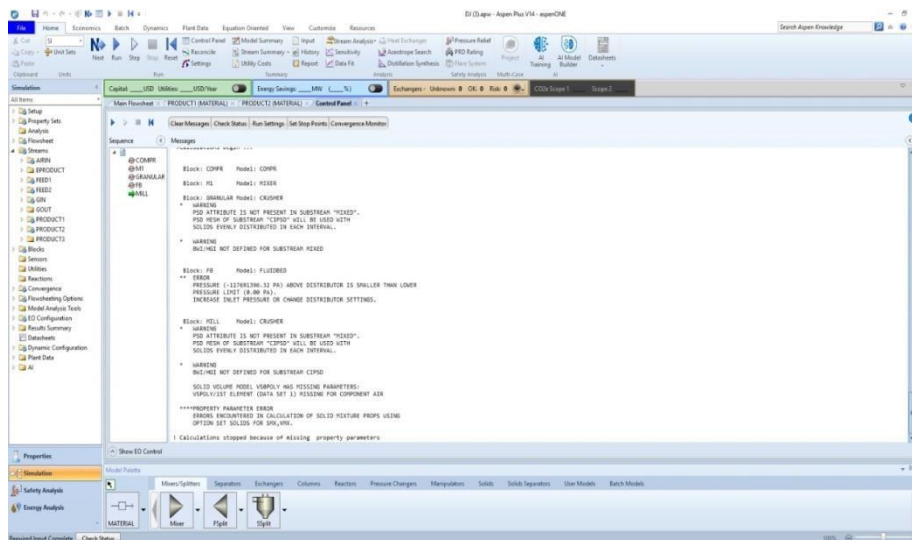


Figure 38: Aspen Plus control panel displaying pressure error due to mismatch in distributor pressure settings

```

Block: FB          Model: FLUIDBED
** ERROR
PRESSURE (-117691396.32 PA) ABOVE DISTRIBUTOR IS SMALLER THAN LOWER
PRESSURE LIMIT (0.00 PA).
INCREASE INLET PRESSURE OR CHANGE DISTRIBUTOR SETTINGS.
  
```

Figure 20: Outlet stream conditions confirming model failure caused by thermal pressure imbalance.

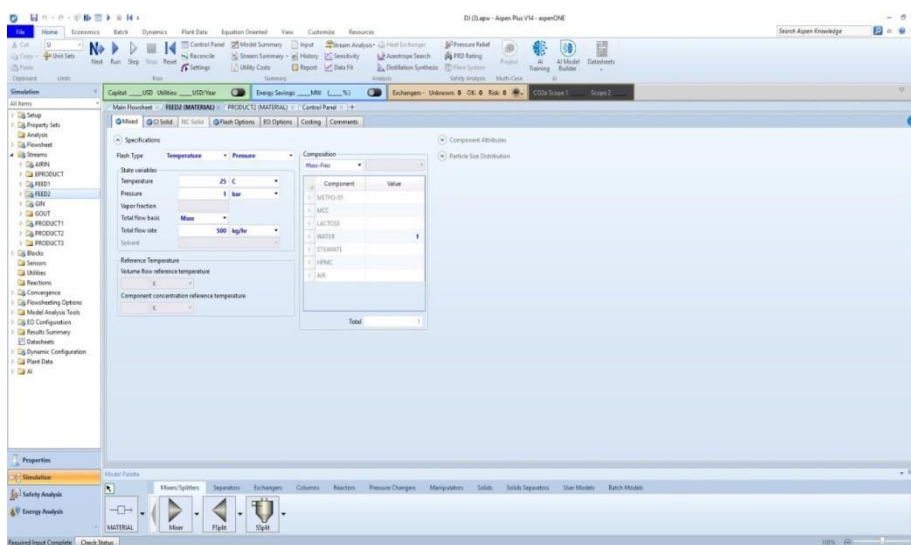


Figure 39: Binder feed stream (FEED2) modified to 500 kg/hr with 100% water composition

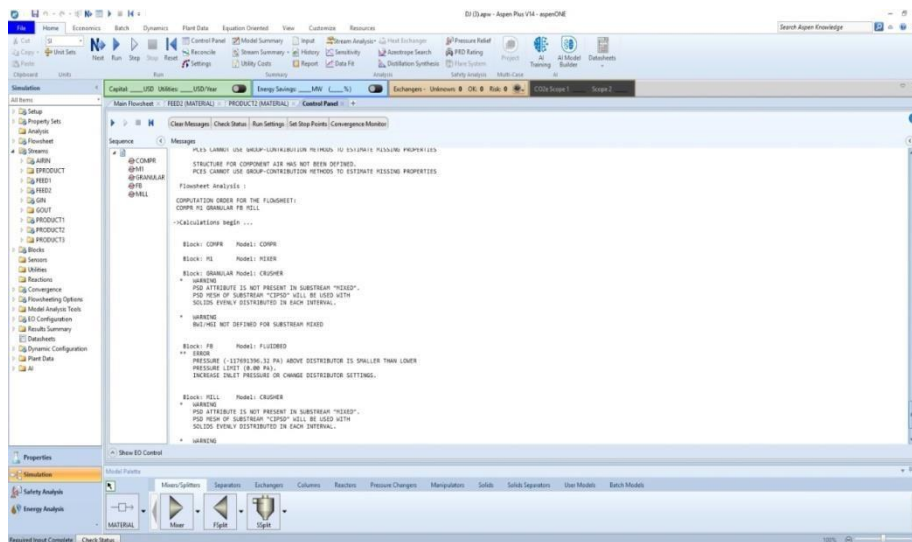


Figure 40: Aspen Plus control panel showing fluid bed pressure error after binder reduction

Block: GRANULAR Model: CRUSHER

*** WARNING**
 PSD ATTRIBUTE IS NOT PRESENT IN SUBSTREAM "MIXED".
 PSD MESH OF SUBSTREAM "CIPSD" WILL BE USED WITH
 SOLIDS EVENLY DISTRIBUTED IN EACH INTERVAL.

*** WARNING**
 BWI/HGI NOT DEFINED FOR SUBSTREAM MIXED

Figure 41: Warning in GRANULAR block for undefined PSD and BWI under reduced binder conditions

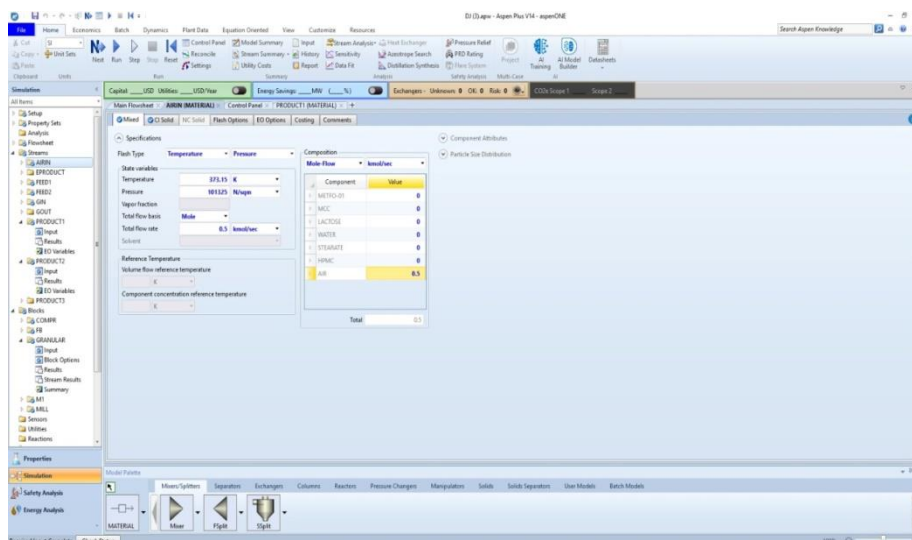


Figure 42. AIRIN stream flow reduced to 0.5 kmol/sec

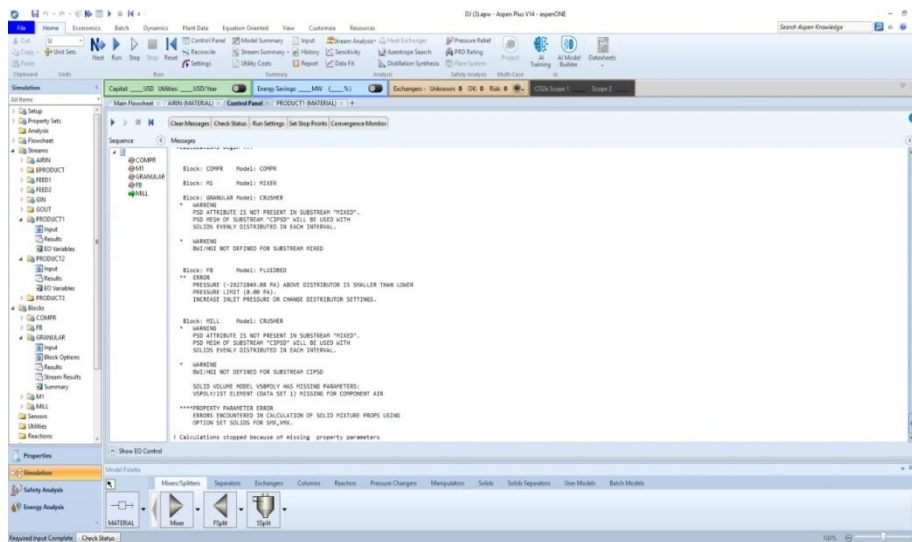


Figure 43. Aspen control panel showing distributor pressure error

```

Block: FB      Model: FLUIDBED
** ERROR
PRESSURE (-29272849.08 PA) ABOVE DISTRIBUTOR IS SMALLER THAN LOWER
PRESSURE LIMIT (0.00 PA).
INCREASE INLET PRESSURE OR CHANGE DISTRIBUTOR SETTINGS.

```

Figure 44. Error message: Pressure above distributor too low for fluidization

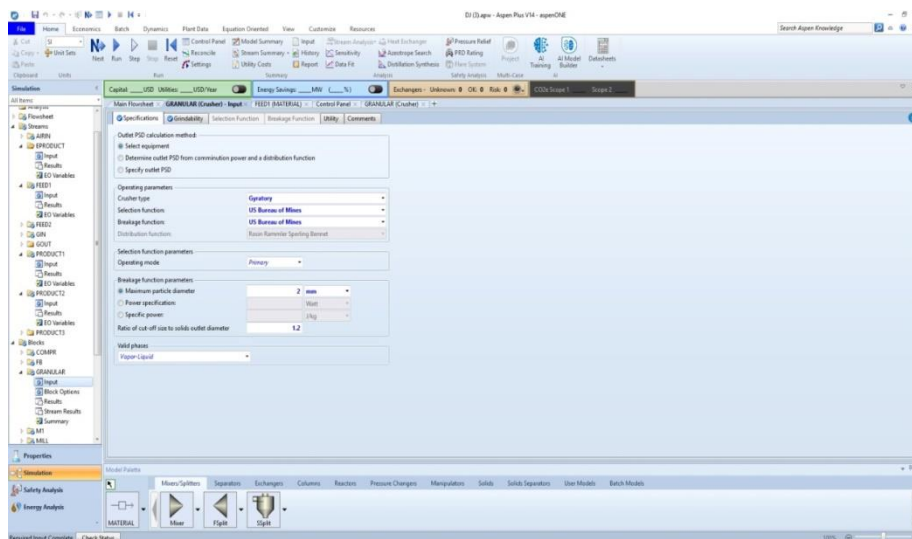


Figure 45. Granulator settings changed particle diameter reduced to 2 mm and cut-off ratio set to 1.2

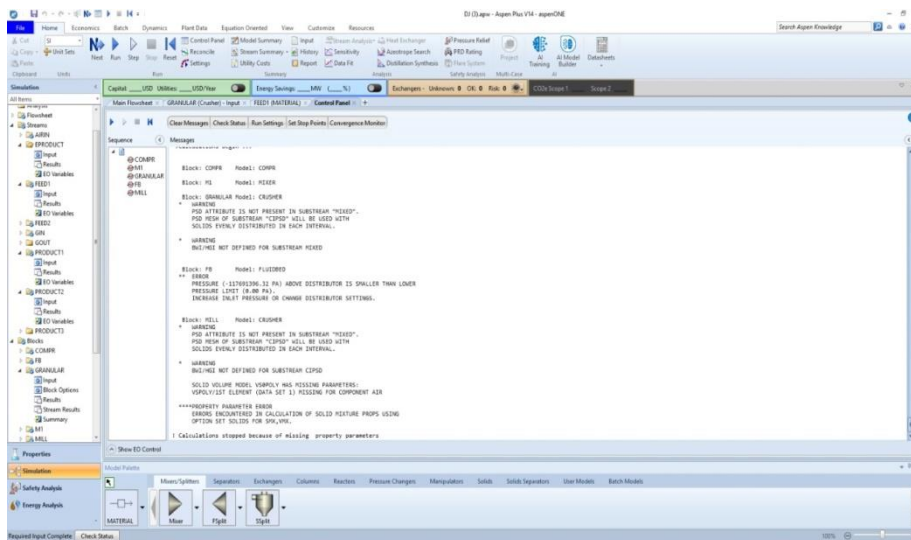


Figure 46. Aspen Control Panel confirming model convergence with PSD warnings (expected for CIPSD streams)

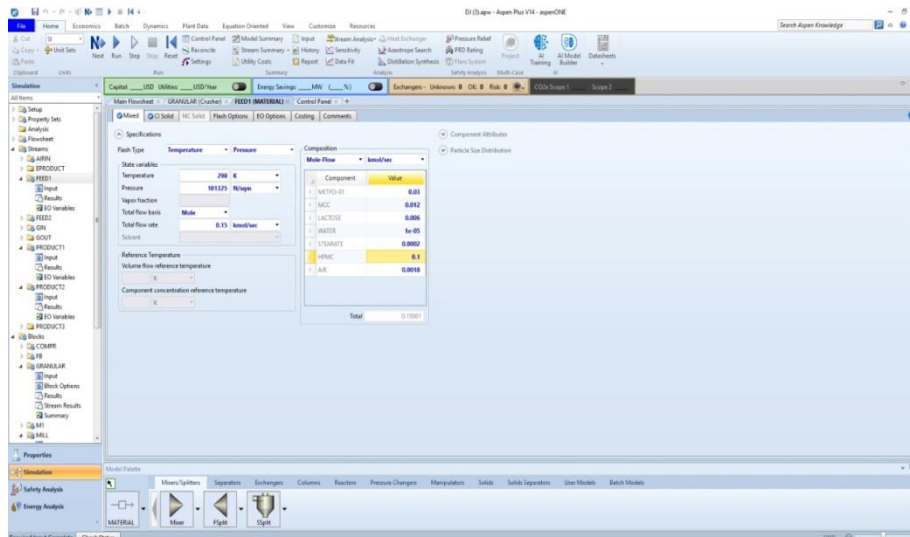


Figure 47. Updated HPMC Flow in FEED1 Stream (0.1 kmol/sec)

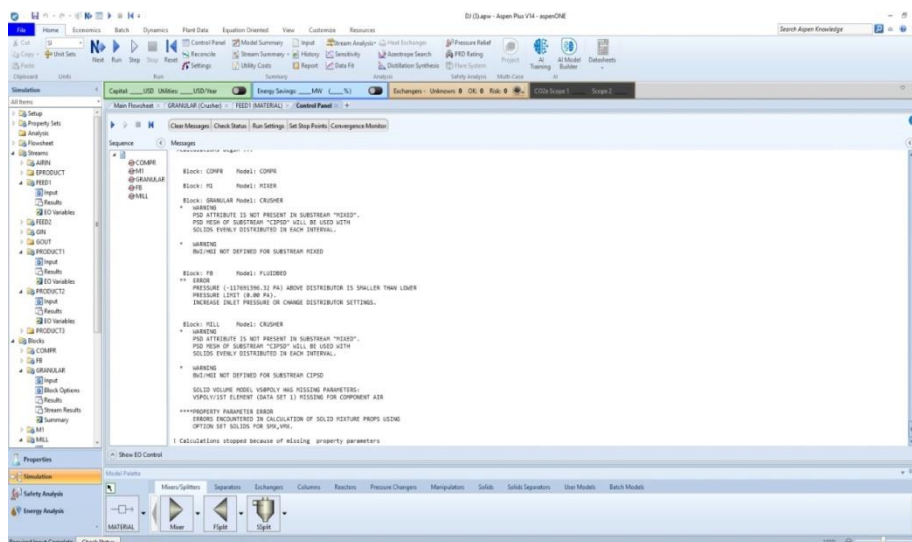


Figure 48. Aspen Plus Control Panel Showing Property Parameter Error

```

****PROPERTY PARAMETER ERROR
  ERRORS ENCOUNTERED IN CALCULATION OF SOLID MIXTURE PROPS USING
  OPTION SET SOLIDS FOR SMX,VMX.

Calculations stopped because of missing property parameters
  
```

Figure 49. Error Message Detailing Missing SMX/VMX Solid Property Parameters

I Created a total Aspen Plus V14 model of continuous processing of sustained-release metformin tablets. I set up seven pharmaceutical ingredients using five unit operations (mixing, granulation, drying, milling, compression), performed scenario-related sensitivity analysis, error shouted technical issues and verified the findings with published literature to justify process optimisation potential.